MAUVE–MUSE: When Metallicity Follows or Fights Star Formation—A Mass-Dependent Inversion in Virgo Galaxies

Rongjun Huang (黄镕钧)[1,★] Luca Cortese[1], Barbara Catinella[1], Luke J. M. Davies[1], Toby Brown[2], Andrei Ristea[3,4], Alessandro Boselli[5], Andrew J. Battisti[1,6], Vicente Villanueva[7,8], Kristine Spekkens[9], Daniel A. Dale[10], Sabine Thater[11], Amirnezam Amiri[12,13], Sara L. Ellison[14]

[1]International Centre for Radio Astronomy Research, University of Western Australia, 7 Fairway, Crawley, 6009, Western Australia, Australia
[2]National Research Council of Canada, Herzberg Astronomy and Astrophysics Research Centre, 5071 W. Saanich Rd., Victoria, BC, V9E 2E7, Canada
[3]Centre for Astrophysics and Supercomputing, Swinburne University of Technology, John St, Hawthorn, 3122, Victoria, Australia
[4]ARC Centre of Excellence in Optical Microcombs for Breakthrough Science (COMBS), Australia
[5]Aix-Marseille Université, CNRS, CNES, LAM, Marseille, France
[6]Research School of Astronomy and Astrophysics, Australian National University, Cotter Road, Weston Creek, ACT, 2611, Australia
[7]Instituto de Estudios Astrofísicos, Facultad de Ingeniería y Ciencias, Universidad Diego Portales, Av. Ejército Libertador 441, 8370191 Santiago, Chile
[8]Millennium Nucleus for Galaxies, MINGAL
[9]Department of Physics, Engineering Physics and Astronomy, Queen's University, Kingston, ON, K7L 3N6, Canada
[10]Department of Physics & Astronomy, University of Wyoming, Laramie, WY 82071, USA
[11]Department of Astrophysics, University of Vienna, T¨urkenschanzstrasse 17, 1180 Wien, Austria
[12]School of Astronomy, Institute for research in fundamental sciences (IPM), Tehran, P.O. Box 19395-5531, Iran
[13]Department of Physics, University of Arkansas, 226 Physics Building, 825 West Dickson Street, Fayetteville, AR 72701, USA
[14]Department of Physics & Astronomy, University of Victoria, Finnerty Road, Victoria, BC, V8P 1A1, Canada

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ABSTRACT

Although globally-integrated studies often find that, at fixed stellar mass, high star formation rate (SFR) galaxies are relatively metal-poor while lower-SFR systems are more metal-rich, the corresponding coupling between gas-phase metallicity ($Z_{\rm gas}$) and star formation on sub-galactic scales remains poorly constrained. In this study, we analyse 14 Virgo spiral galaxies from the MAUVE–MUSE survey to revisit the resolved mass–metallicity relation (rMZR) and its secondary dependence on star formation rate surface density ($\Sigma_{\rm SFR}$) at $\sim$ 100 pc scales. We construct co-spatial maps of stellar mass surface density (Σ_*), $\Sigma_{\rm SFR}$ and gas-phase oxygen abundance and verify our results across several strong-line metallicity diagnostics. MAUVE–MUSE galaxies follow a standard rMZR, but once binned in Σ_*, we find a clear mass-dependent inversion in the $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ relation: low-Σ_* bins show an anti-correlation between $Z_{\rm gas}$ and $\Sigma_{\rm SFR}$, whereas high-Σ_* regions show a positive correlation, with an inversion point at $\log_{10}(\Sigma_*/M_\odot\,{\rm kpc}^{-2}) \simeq 7.5$–8.0. We find that both correlated and anti-correlated H II regions coexist within the same discs, and that the observed mass dependence emerges only when grouping H II spaxels by Σ_*. To interpret this behaviour, we develop a spatially-resolved gas–regulator model and show that the observed correlation and anti-correlation between $Z_{\rm gas}$ and $\Sigma_{\rm SFR}$ arise from the competition between star-formation-driven and gas-supply-driven variability: the sign of the local $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ relation is set by which of these dominates. This framework can be naturally extrapolated to the integrated scenario, providing a unified explanation for both the resolved and global $Z_{\rm gas}$–SFR relation.

Key words: galaxies: abundances — galaxies: ISM — galaxies: star formation — galaxies: evolution — ISM: abundances

1 Introduction

The global stellar mass–gas-phase metallicity relation (MZR) (e.g. Lequeux et al. 1979; Tremonti et al. 2004; Erb et al. 2006; Maiolino et al. 2008; Zahid et al. 2014; Curti et al. 2020; Sanders et al. 2021), demonstrates that gas-phase metallicity ($Z_{\rm gas}$, also commonly referred to as gas-phase oxygen abundance) increases with galaxy stellar mass (M_*), such that more massive galaxies host a more metal-enriched interstellar medium (ISM). Physically, this trend is commonly interpreted as the outcome of mass-dependent chemical regulation: in the low-mass regime the near-linear trend is a consequence of the integrated star-formation history, whereas the high-mass flattening is attributed to stronger regulation by outflows and/or saturation toward the maximum oxygen yield (see Maiolino & Mannucci 2019 for a review). While this global relation provides an essential benchmark for obtaining a comprehensive picture of chemical

★ E-mail: Rongjun.Huang@icrar.org, Astro@Rongjun-Huang.com

evolution studies, the advent of integral-field spectroscopy (IFS) has shifted attention toward understanding how metallicity is regulated within galaxies on a resolved basis. By resolving discs into individual star-forming spaxels, IFS surveys now enable direct measurement of local scaling relations between gas-phase oxygen abundance, and local stellar and star formation properties (see Sánchez 2020 for a review).

The Calar Alto Legacy Integral Field Area (CALIFA) survey (Sánchez et al. 2013, 2017; Cresci et al. 2019) revealed that the global MZR has a local counterpart: on kpc-scale spatial resolution, gas-phase oxygen abundance correlates tightly with the local stellar mass surface density (Σ_*). This resolved mass–metallicity relation (rMZR) was subsequently confirmed using ~~the~~ much larger and more diverse samples including the Mapping Nearby Galaxies at APO (MaNGA; Barrera-Ballesteros et al. 2016; Baker et al. 2023) and the Sydney-AAO Multi-object Integral-field spectrograph (SAMI; Sánchez et al. 2019) surveys, establishing the rMZR as a ubiquitous feature of star-forming discs on kpc scales in the local Universe. More recently, IFS surveys based on Very Large Telescope/Multi-Unit Spectroscopic Explorer (VLT/MUSE) have extended this resolved scaling relation to intermediate redshift ($0.1 \lesssim z \lesssim 1$) while retaining comparable spatial resolution (~kpc scale). In particular, the MUSE–Wide survey demonstrated the existence of an rMZR at $z \sim 0.26$ (Yao et al. 2022), with a shape similar to that at $z \sim 0$ but a modestly lower normalization, consistent with our models of chemical evolution over cosmic time. Likewise, the Middle Ages Galaxy Properties with Integral Field Spectroscopy (MAGPI) survey confirmed an rMZR in galaxies at $z \sim 0.3$ on a similar (kpc-scale) spatial resolution (Koller et al. 2024). Additionally, with the James Webb Space Telescope/Near Infrared Spectrograph (JWST/NIRSpec), the ALPINE–CRISTAL–JWST programme is now extending rMZR studies to representative main-sequence galaxies at $z \simeq 4$–6 at resolutions of a few hundred pc, with $Z_{\rm gas}$ showing a stronger dependence on $\Sigma_{\rm SFR}$ than those in local universe (Fujimoto et al. 2025). Taken together, these IFS studies indicate that gas-phase oxygen abundance correlates more tightly with local stellar mass surface density than with galactocentric radius alone (e.g. Barrera-Ballesteros et al. 2016; Baker et al. 2023): at fixed radius, spaxels with higher Σ_* tend to be more metal-rich, and a similar Σ_*–$Z_{\rm gas}$ relation is recovered in galaxies with very different sizes and radial profiles. In this sense, the rMZR appears to reflect local physics, even though it is closely linked to the underlying radial structure of galactic discs.

On global scales, there is now broad consensus that gas-phase metallicity depends on both stellar mass and star formation rate (SFR): the classical MZR is usually interpreted as a projection of a more fundamental M_*–$Z_{\rm gas}$–SFR surface/plane, often referred to as the Fundamental Metallicity Relation (FMR; e.g. Ellison et al. 2008; Mannucci et al. 2010; Lara-López et al. 2010; Yates et al. 2012; Zahid et al. 2013; Bothwell et al. 2013; Peng & Maiolino 2014; Bothwell et al. 2016; Gao et al. 2018; Cresci et al. 2019; Curti et al. 2020; Pallottini et al. 2025). In most of these studies, galaxies with higher SFR at fixed M_* tend to be more metal-poor, i.e. $Z_{\rm gas}$ and SFR are anti-correlated, a behaviour often interpreted as the signature of metal-poor gas accretion diluting the ISM and fuelling star-formation. This interpretation naturally raises the question of whether an analogous three-parameter relation exists on resolved scales, i.e. whether the rMZR carries a meaningful secondary dependence on local star formation rate surface density ($\Sigma_{\rm SFR}$). Several works based on large, predominantly star-forming disc samples report that once the primary Σ_* dependence is removed, residual trends with $\Sigma_{\rm SFR}$ are weak or absent on kpc physical scales, suggesting that local metallicity primarily traces longer-term mass assembly and recycling (e.g. Sánchez et al. 2013; Yao et al. 2022). However, recent studies find a statistically significant $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ (anti-)correlation and motivate a spatially-resolved analogue of the Fundamental Metallicity Relation (i.e., rFMR), wherein metallicity at fixed Σ_* depends on $\Sigma_{\rm SFR}$ (Sánchez-Menguiano et al. 2019; Baker et al. 2023; Koller et al. 2024). Additionally, the MAGPI data presents a clear signal that the *sign* of this correlation can reverse with Σ_*: low-Σ_* regions show an anti-correlation between $Z_{\rm gas}$ and $\Sigma_{\rm SFR}$, while high-Σ_* regions exhibit a flat or positive relation (Koller et al. 2024). Such global and resolved studies therefore suggest that any FMR- or rFMR-like behaviour might not be universal but instead depend on local physical regime, calibration choice, and whether variability in metallicity is dominated by star-formation-driven enrichment or by gas-supply-driven dilution.

The potential flip between the $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ correlation and anti-correlation suggests a link between metallicity variations and the gas-star formation cycle. Interpreting such mass-dependent sign changes requires a framework that links metallicity variations to the gas–star formation cycle. A widely used description of this link is provided by the "gas–regulator" or "bathtub" models, in which the gas content and metal budget of a galaxy (or region of a galaxy) are governed by the balance between gas inflow, star formation, and outflows (e.g. Davé et al. 2012; Lilly et al. 2013; Peng & Maiolino 2014; Forbes et al. 2014; Belfiore et al. 2019; Sharda et al. 2021). In their galaxy-integrated form, these models predict that the gas-phase metallicity tends toward an effective equilibrium set by the metal yield, the metallicity of the accreted gas, and the interplay between star formation and inflow rate, naturally reproducing both the shape of the global MZR and its commonly observed anti-correlation with SFR. A more explicit connection between time-dependent star formation and metallicity has been established by Wang & Lilly (2021, hereafter WL21), who treated individual "gas–regulator units" (from galaxy-wide down to ~ 100 pc scales) as reservoirs subject to time-variable inflow rate and star formation efficiency (SFE). Within this framework, the residual correlation between metallicity and SFR at fixed stellar mass (or gravitational potential) is set by which source of variability dominates: when fluctuations in SFE drive changes in the SFR at roughly fixed inflow rate, metallicity and SFR are predicted to have a positive correlation, whereas strongly time-variable inflow at nearly constant SFE produces metal dilution and thus a negative correlation between metallicity and SFR. However, testing such analytical frameworks on local scales requires spatially-resolved data that simultaneously (i) resolve individual H II regions on ~ 100 pc scales and (ii) span a broad dynamic range in Σ_*, so that both SFR-driven and inflow-driven regimes can be probed within and across galaxies.

In this work, we use observations of the first 14 Virgo Cluster spirals observed as part of the Multiphase Astrophysics to Unveil the Virgo Environment with MUSE (MAUVE–MUSE) survey (Catinella et al. 2025) to revisit the rMZR and its secondary dependence on $\Sigma_{\rm SFR}$ at ~ 100 pc resolution. We show that the Virgo cluster spirals sample exhibits a robust mass-dependent inversion in both the $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ relation and in the corresponding residual space ($\Delta Z_{\rm gas}$–$\Delta\Sigma_{\rm SFR}$ relation), where $\Delta Z_{\rm gas}$ and $\Delta\Sigma_{\rm SFR}$ denote offsets at fixed Σ_* from the mean resolved trends. This behaviour is consistent with previous indications of a sign reversal, but extending them to a cluster sample at ~ 100 pc resolution for the first time. To characterise the spatial heterogeneity of this coupling, we introduce a local bivariate Moran-like statistic (Moran 1950; Anselin 1995) that quantifies, around each spaxel, whether nearby regions tend to show same-sign or opposite-sign pairs of ($\Delta Z_{\rm gas}$, $\Delta\Sigma_{\rm SFR}$), thereby identifying spatially coherent zones of concordant or discordant. Finally, we develop a spatially-resolved regulator model to interpret these trends in our data. The results demonstrate that the observed mass-dependent sign flip in the $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ relation is governed by the relative amplitude of local star-formation versus gas-supply variability.

This paper is organized as follows: Section 2 introduces the data and methods used in this study, Section 3 outlines our results, Section 4 presents the spatially-resolved framework, its interpretation and the extrapolation to a global integrated scenario, Section 5 discusses the robustness of our findings across different metallicity calibrations, star-forming spaxel selections, and the special case of the metallicity outlier NGC 4383, and Section 6 outlines our conclusions. Throughout this work, we adopt a distance of 16.5 Mpc for Virgo Cluster members (Mei et al. 2007), aligning with previous and ongoing MAUVE studies; the flat Lambda Cold Dark Matter (ΛCDM) model is adopted by assuming the Hubble constant is H_0 = 70km/s/Mpc and the mass density of the Universe is $\Omega_{\rm m,0} = 0.3$.

2 Data & Methods

2.1 MAUVE–MUSE Data

The data analysed in this work are drawn from the internal release v2 (December 2024) of the MAUVE–MUSE survey. Designed to investigate how the cluster environment shapes the gas–star formation cycle, MAUVE targets 40 late-type Virgo Cluster galaxies using a suite of multi-wavelength facilities (Catinella et al. 2025). For each target, we use the fully reduced MUSE datacubes produced with `pymusepipe` v2.28.2 and `esorex` 3.13.6, following closely the reduction, sky subtraction, and flux-calibration procedures described by Emsellem et al. (2022). The data for each galaxy is delivered as a science-ready, fully-reduced FITS file on the native 0.″2 pixel grid over the full 4750–9350 Å spectral range of VLT/MUSE.

Stellar-continuum and emission-line maps used in this paper are derived from these cubes using the `nGIST` pipeline (Fraser-McKelvie et al. 2025), a `pPXF`-based (Cappellari & Emsellem 2004; Cappellari 2016, 2023) wrapper optimised for MAUVE–MUSE data. Briefly, we first construct an adaptively binned representation of each cube by applying the Voronoi algorithm of Cappellari & Emsellem (2004) to the stellar continuum in the 4800–7000 Å spectral window; spaxels are then aggregated until the bin-level S/N, computed by summing signal and adding noise in quadrature, reaches a target of S/N ≥ 40. The resulting Voronoi tessellation is then adopted as the common spatial grid for all `nGIST` products. This approach ensures strict one-to-one correspondence between all the following derived properties without any additional resampling or smoothing. The MAUVE–MUSE observations have a typical seeing full width at half maximum (FWHM) of $\sim 1''$, corresponding to a physical scale of $\sim$ 80 pc at the Virgo Cluster's distance, so that the effective angular resolution is set by the point spread function (PSF) rather than by the MUSE sampling pixel scale of 0.″2 pixel^{-1}. The S/N ≥ 40 Voronoi binning on the stellar continuum typically groups a few up to several tens of spaxels per bin with 95% containing fewer than 40 spaxels, yielding characteristic bin sizes comparable to the seeing footprint and hence also of order $\sim$ 100 pc. Throughout this work, we therefore refer to our spatially-resolved measurements as probing $\sim$ 100 pc scales.

On this Voronoi grid of each galaxy, the stellar continuum and kinematics in each bin were fit with `pPXF` using MILES† SSP templates (Vazdekis et al. 2010), after convolving the templates with a wavelength-dependent Gaussian MUSE line–spread function (LSF), parameterised following Equation 8 of Bacon et al. (2017) (see also Emsellem et al. 2022). Multiplicative Legendre polynomials are included to account for residual relative flux calibration offsets. After subtracting the best-fitting stellar model, Gaussian profiles are fitted simultaneously to the main nebular emission lines (e.g. Hα, Hβ, [O III], [N II], [S II]), yielding maps of observed line fluxes, velocities, and velocity dispersions with associated uncertainties.

While MAUVE–MUSE observations are still in progress, a total of 14 galaxies have been fully observed, reduced and therefore analyzed in this paper (see Catinella et al. 2025 for full details of the MAUVE–MUSE sample): IC 3392, NGC 4064, NGC 4192, NGC 4293, NGC 4298, NGC 4330, NGC 4383, NGC 4396, NGC 4419, NGC 4457, NGC 4501, NGC 4522, NGC 4694, and NGC 4698.

2.2 Stellar Mass Surface Density

We first derive spatially-resolved stellar mass surface densities from the MAUVE–MUSE datacubes and the associated `nGIST` value-added products. For each galaxy, we start from the `nGIST` star-formation-history (SFH) products, which provide, for every Voronoi bin, a vector of SSP template weights (i.e. the relative light contributions of each stellar metallicity-age component in the spectral fit) and the corresponding SSP parameters grid in stellar metallicity and age, $([\mathrm{M/H}], \log_{10}(t))$. We then adopt the MILES-based SSP (Vazdekis et al. 2010) predictions based on BaSTI isochrone grids (Pietrinferni et al. 2004) with a Chabrier (2003, hereafter Chabrier) initial mass function (IMF), using their tabulated Johnson R-band mass-to-light ratios ($(M/L)_R$) as a function of $([\mathrm{M/H}], \log_{10} t)$. We use the R band as the MILES SSP templates provide the (M/L) in Johnson bands, and the R band is fully and comfortably covered by MUSE, reducing band-edge systematics relative to V and I bands. The SFH weights for each Voronoi bin are then combined with this $(M/L)_R$ grid to compute a light-weighted R-band mass-to-light ratio,

$$(M/L)_R^{\rm bin} = \sum_k w_k \, (M/L)_{R,k}, \quad (1)$$

where w_k and $(M/L)_{R,k}$ are the SFH weight and R-band mass-to-light ratio of the k-th SSP component in the bin based on variable stellar metallicities and ages. This procedure yields a resolved map of $(M/L)_R^{\rm bin}$ defined over all Voronoi zones.

To obtain the corresponding R-band luminosity per bin, we construct the observed spectral energy distribution, F_λ, and convolve it with the `bessell-R` filter response using the `speclite` package (Kirkby et al. 2023). This yields an AB R-band magnitude, m_R, and flux (in nanomaggies) per spaxel. We then collapse these spaxel-level fluxes onto the Voronoi tessellation by averaging, only over spaxels assigned to that bin, while preserving the survey's native masking of invalid pixels. The resulting bin-averaged AB magnitudes are corrected for Galactic extinction using the `nGIST` $E(B-V)$ map and a Calzetti et al. (2000, hereafter Calzetti) attenuation curve. Corrected apparent magnitudes ($m_R^{\rm corr}$) are converted into absolute magnitudes via the distance modulus corresponding to the adopted Virgo Cluster distance of 16.5 Mpc, and hence into R-band luminosities (L_R) in solar units using the solar R-band absolute magnitude, $M_{R,\odot} = 4.61$ (Willmer 2018).

The stellar mass in each Voronoi bin is then obtained as

$$M_*^{\rm bin} = L_R^{\rm bin} \times (M/L)_R^{\rm bin}. \quad (2)$$

Dividing $M_*^{\rm bin}$ by the corresponding projected bin area yields a map of projected stellar mass surface density, Σ_* (in $M_\odot$ kpc^{-2}).

Finally, we correct observed surface densities for disc inclination,

† MILES website: https://research.iac.es/proyecto/miles/

adopting inclination angles (i) from the Virgo Environment Traced in CO (VERTICO) Survey (Brown et al. 2021), where $i = 0°$ is face-on and b/a is the axis ratio in projection. Assuming an oblate disc geometry with intrinsic thickness, $q_0 = 0.2$, we compute the inclination correction factor for each galaxy:

$$\frac{b}{a} = \sqrt{(1 - q_0^2)\cos^2 i + q_0^2}, \tag{3}$$

which encodes the observed flattening of a disc of finite thickness. The stellar mass surface density and the star formation rate surface density in the following Section 2.3 are then multiplied by this b/a factor to correct from the projected to the intrinsic face-on surface densities (i.e. to remove foreshortening). This correction addresses geometry only and does not explicitly model inclination-dependent attenuation.

While all spectral analyses (including the stellar population fitting here and the nebular emission fitting below) are performed on the Voronoi-binned spectra to ensure a minimum continuum $S/N \geq 40$, we reassign the derived properties of each bin back to its constituent spaxels to create resolved maps and undertake statistical analysis. This procedure preserves the survey's native spatial sampling of $0.''2\,\mathrm{pixel}^{-1}$, effectively weighting each Voronoi bin by its projected area. Consequently, unless otherwise noted, the term "spaxel" in the following contexts refers to these grid elements on the native scale, which carry the robust physical values derived from their parent Voronoi bin. Additionally, in agreement with the findings of Koller et al. (2024), we found that adopting a bin-based analysis does not significantly change the results presented in this work compared to the current spaxel-based approach.

2.3 Star Formation Rate Surface Density

Nebular dust attenuation is quantified from the Balmer Decrement (BD) measured on the same Voronoi grid as all other MAUVE–MUSE products. For each spaxel, we start from the `nGIST` Hα and Hβ flux and error maps (F_{line} and σ_{line}, where line = Hα and Hβ) and impose a basic quality-control cut requiring $F_{\mathrm{line}}/\sigma_{\mathrm{line}} \geq 3$, together with a minimum flux threshold of $F_{\mathrm{line}} \geq 20\times10^{-20}\,\mathrm{erg\,s^{-1}\,cm^{-2}}$ per spaxel in the native MAUVE–MUSE units. Spaxels that fail this detection criterion for either Balmer line are retained for the construction of SFR maps, but their fluxes are set to max(threshold, F_{line}) (effectively assuming zero extinction) to prevent artifacts such as unphysical extinction corrections. These low-S/N spaxels will be excluded from our star-forming spaxel selection later, so this choice has no impact on our results. The Balmer Decrement is defined as $\mathrm{BD} = F_{\mathrm{H}\alpha}/F_{\mathrm{H}\beta}$ (using observed fluxes) and we impose a floor corresponding to the Case B recombination value at $\mathrm{BD} = 2.86$, appropriate for electron temperature and number density at $T_e = 10^4$ K and $n_e = 100\,\mathrm{cm^{-3}}$ (Osterbrock & Ferland 2006), i.e. any measured $\mathrm{BD} < 2.86$ is set to 2.86 to avoid unphysical negative colour excesses. The corresponding colour excess is obtained using the following form (Osterbrock & Ferland 2006):

$$E(B-V)_{\mathrm{gas}} = \frac{2.5}{k(\mathrm{H}\beta) - k(\mathrm{H}\alpha)} \log_{10}\left(\frac{\mathrm{BD}}{2.86}\right), \tag{4}$$

where $k(\lambda)$ is the Milky Way-like extinction curve with the effective obscuration at V-band $R_V = 3.1$ (Cardelli et al. 1989). All strong nebular lines entering the subsequent analysis are then dereddened with this $E(B-V)_{\mathrm{gas}}$ via

$$F_\lambda^{\mathrm{corr}} = F_\lambda\, 10^{0.4\,E(B-V)_{\mathrm{gas}}\,k(\lambda)}. \tag{5}$$

Using these dust-corrected fluxes we construct the standard [N II] and [S II] Baldwin et al. (1981, hereafter BPT) ratios, namely $F_{[\mathrm{N\,II}]\lambda6583}/F_{\mathrm{H}\alpha}$, $F_{[\mathrm{S\,II}](\lambda6716+\lambda6731)}/F_{\mathrm{H}\alpha}$, and $F_{[\mathrm{O\,III}]\lambda5007}/F_{\mathrm{H}\beta}$, propagating flux-to-error ratios into the corresponding line-ratio uncertainties (see Figure 1). Each spaxel is classified on both the [N II] and [S II] BPT diagrams according to the Kewley et al. (2006) demarcation curves, and we require that the central value and its $\pm1\sigma$ error bars $(\log_{10}([\mathrm{O\,III}]/H\beta), \log_{10}(\mathrm{line}/\mathrm{H}\alpha))$ all remain within the same BPT region. Only spaxels that satisfy this criterion and the quality-control cuts in all lines entering a given diagram are assigned a secure class (6.40% spaxels that straddle a demarcation curve at the $\pm1\sigma$ level are excluded).

We define "star-forming" spaxels (hereafter **HII** mask) as those that lie in the H II regime of both [N II] and [S II] BPT diagrams and whose $\pm1\sigma$ error bars also remain entirely within the H II regime on both diagrams, with robust detections in all lines entering the classification. This yields a clean sample of classical H II regions dominated by O/B stars' photonization with minimal contamination from AGN or LINER-like excitation (Lacerda et al. 2020). All SFR and metallicity maps in this work are therefore computed exclusively using this **HII** mask, and all subsequent resolved analyses implicitly refer to these H II-classified star-forming regions. Across the sample, the conservative **HII** selection retains ~31.44% of the total valid spaxels used in this work (with substantial galaxy-to-galaxy variation from 3.71% to 48.38%). Additionally, in Section 5.2, we perform a robustness test by relaxing this conservative "star-forming" restriction and repeating the analysis with an expanded **HII+Comp** mask that also includes spaxels classified as composite on both BPT diagrams.

Star formation rate maps are constructed from the dust-corrected Hα flux. After applying the Balmer-decrement attenuation correction described above, we apply the Calzetti et al. (2007) calibration, rescaled to a Chabrier IMF by Kennicutt & Evans (2012), to obtain

$$\mathrm{SFR}\,[M_\odot\,\mathrm{yr^{-1}}] = 4.98\times10^{-42}\,L_{\mathrm{H}\alpha}^{\mathrm{corr}}\,[\mathrm{erg\,s^{-1}}]. \tag{6}$$

Similarly, we take the projected surface density of SFR and account for the inclination by Equation 3, obtaining corrected Σ_{SFR}.

2.4 Gas-phase Metallicity

Gas-phase oxygen abundances‡ are inferred from the dust-corrected emission-line maps, restricted to the **HII** spaxels defined in Section 2.3. Using only spaxels that meet the S/N and star-forming selections, we focus on a subset of four widely used strong-line prescriptions that are robustly applicable across the MAUVE–MUSE sample: the N2S2-Hα calibration of Dopita et al. (2016, hereafter D16), the S-calibration of Pilyugin & Grebel (2016, hereafter Scal-PG16), the O3N2 index calibration of Marino et al. (2013, hereafter O3N2-M13), and the O3N2 calibration of Curti et al. (2020, hereafter O3N2-C20). The line ratios entering these prescriptions are listed in Table 1. Similar to Watts et al. (2024), we reconstruct the total [N II] $\lambda\lambda6548,6583$ and [O III] $\lambda\lambda4959,5007$ fluxes by scaling the measured [N II] $\lambda6583$ and [O III] $\lambda5007$ fluxes by fixed factors of 1.34 and 1.33, respectively, given by theoretical line ratios in quantum mechanics (Storey & Zeippen 2000). All strong-line calibrations are applied within the recommended range $7.63 \leq [\mathrm{O/H}] \leq 9.23$ by Kewley et al. (2019).

The D16 prescription combines the $N2S2$ and [N II]/Hα ratios into

‡ Throughout this paper, we use $[\mathrm{O/H}] \equiv 12 + \log_{10}(\mathrm{O/H})$ to denote the gas-phase oxygen abundance, with a surface solar abundance of $12 + \log_{10}(\mathrm{O/H}) = 8.69$ (Asplund et al. 2009).

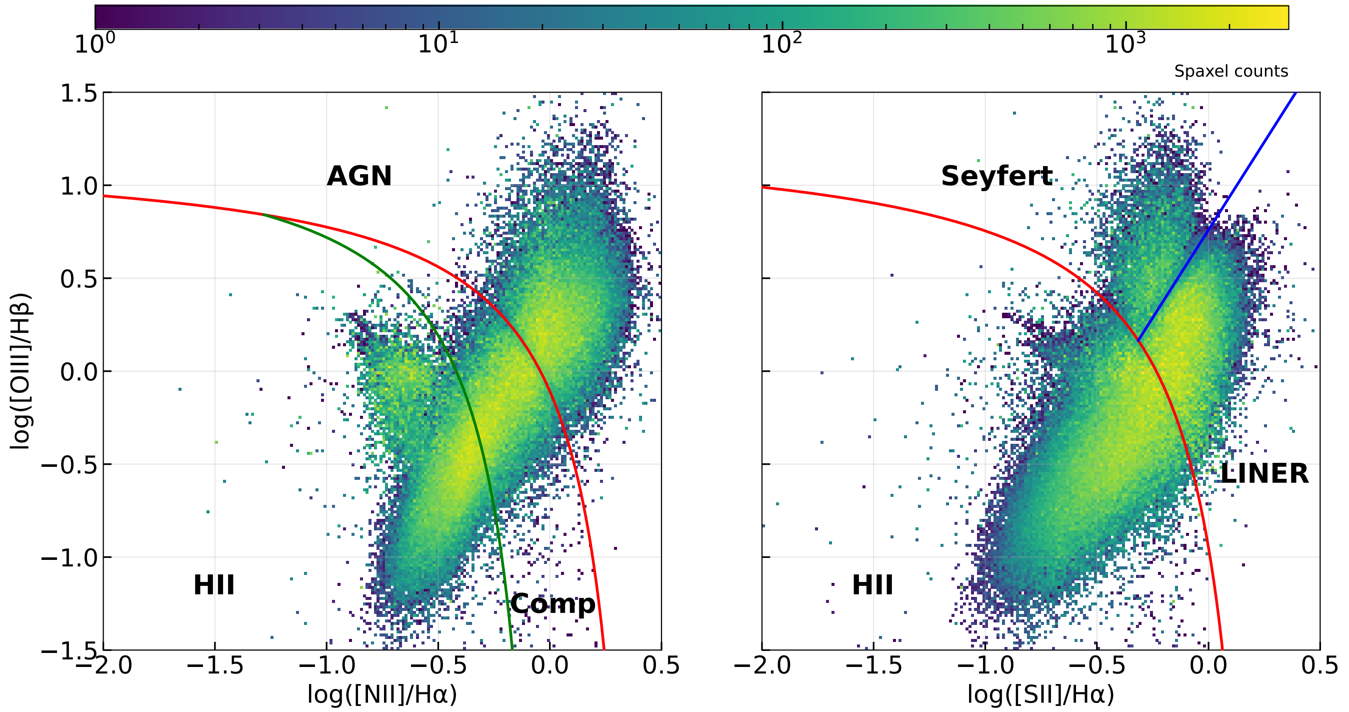

Figure 1. [N II] (left) and [S II] (right) BPT diagrams for "classified" spaxels of all 14 MAUVE–MUSE galaxies. Data points correspond to Voronoi spaxel classified using demarcation lines within 1σ confidence described in Section 2.3, colorcoding by spaxel counts. Red and green curves show the Kewley et al. (2001) and Kauffmann et al. (2003) demarcation lines in the [N II] diagram, while red and blue curves indicate the Kewley et al. (2001) and Kewley et al. (2006) divisions in the [S II] diagram. Spaxels classified as star-forming (**HII** spaxels) require identification as H II regions on both BPT diagrams.

Line ratio	Definition
$N2$	$\frac{F_{\rm [N\,II](\lambda6548+\lambda6584)}}{F_{\rm H\beta}}$
$S2$	$\frac{F_{\rm [S\,II](\lambda6717+\lambda6731)}}{F_{\rm H\beta}}$
$R3$	$\frac{F_{\rm [O\,III](\lambda4959+\lambda5007)}}{F_{\rm H\beta}}$
$O3N2$	$\log_{10}\left(\frac{F_{\rm [O\,III]\lambda5007}/F_{\rm H\beta}}{F_{\rm [N\,II]\lambda6583}/F_{\rm H\alpha}}\right)$
$N2S2$	$\log_{10}\left(\frac{F_{\rm [N\,II]\lambda6583}}{F_{\rm [S\,II](\lambda6716+\lambda6731)}}\right)$
$y_{\rm D16}$	$N2S2 + 0.264\,\log_{10}\left(\frac{F_{\rm [N\,II]\lambda6583}}{F_{\rm H\alpha}}\right)$

Table 1. Definition of key emission-line ratios used in the extinction, SFR, and metallicity analysis, where all fluxes are dust-corrected. Here we adopt $F_{\rm [N\,II](\lambda6548+\lambda6584)} = 1.34 F_{\rm [N\,II]\lambda6584}$ and $F_{\rm [O\,III](\lambda4959+\lambda5007)} = 1.33 F_{\rm [O\,III]\lambda5007}$.

a single abundance indicator. Using the shorthand in Table 1, we obtain the oxygen abundance as

$$[{\rm O/H}]_{\rm D16} = 8.77 + y_{\rm D16} + 0.45(y_{\rm D16} + 0.30)^5. \tag{7}$$

The Scal-PG16 calibration uses the $N2$, $S2$, and $R3$ ratios. We evaluate the branch-dependent polynomial expressions given in Pilyugin & Grebel (2016):

$$[{\rm O/H}]_{\rm PG16} = \begin{cases} 8.424 + 0.030\,\log_{10}(R3/S2) + 0.751\,\log_{10}(N2) \\ \quad +\left[-0.349 + 0.182\,\log_{10}(R3/S2)\right. \\ \quad \left.+0.508\,\log_{10}(N2)\right]\log_{10}(S2), \\ \quad \log_{10}(N2) \geq -0.6; \\ 8.072 + 0.789\,\log_{10}(R3/S2) + 0.726\,\log_{10}(N2) \\ \quad +\left[1.069 - 0.170\,\log_{10}(R3/S2)\right. \\ \quad \left.+0.022\,\log_{10}(N2)\right]\log_{10}(S2), \\ \quad \log_{10}(N2) < -0.6. \end{cases} \tag{8}$$

For the O3N2-M13 calibration, we adopt the O3N2 index defined in Table 1 and use the corresponding abundance relation:

$$[{\rm O/H}]_{\rm O3N2-M13} = 8.533 - 0.214\,O3N2, \tag{9}$$

Finally, we also adopt the O3N2-based calibration of Curti et al. (2020), which uses the same O3N2 index but fits a higher-order polynomial to stacked Sloan Digital Sky Survey (SDSS; Abazajian et al. 2009) spectra. We evaluate the recommended O3N2-C20 functional form spaxel-by-spaxel over the valid range:

$$O3N2 = 0.281 - 4.765\,x - 2.268\,x^2, \tag{10}$$

where $x \equiv [{\rm O/H}]_{\rm O3N2-C20} - 8.69$. In the course of this work, we also implemented the multi-indicator ("combined") strategy advocated by Curti et al. (2020). In this approach, the oxygen abundance is inferred for each spaxel by jointly comparing the observed set of available strong-line ratios ($R3$, $N2$, $S2$, $O3N2$ and $O3S2$; see Table 1 of Curti et al. 2020) to the corresponding empirical calibrations and selecting the best-fit solution by minimizing the χ^2 statistics that accounts for the measurement uncertainties and the intrinsic calibration scatter. While this method yields self-consistent metallicity estimates with formally small uncertainties, we find that it can introduce diagnostic-driven discontinuities within individual galaxies when different ratios dominate in different regions. To ensure internal consistency and straightforward interpretation of spatial trends, we therefore adopt the single $O3N2$ calibration of Curti et al. (2020).

All metallicity prescriptions are evaluated exclusively on the conservative **HII** star-forming mask, on the same Voronoi grid as the stellar-mass and SFR surface-density maps. This guarantees that Σ_*, $\Sigma_{\rm SFR}$, and $12+\log_{10}({\rm O/H})$ are strictly co-spatial for every bin used in the resolved mass–metallicity–SFR analysis presented in Section 3. Throughout this paper, to be consistent, we adopt O3N2-M13 as the standard prescription. We select this indicator to ensure robustness and comparability across studies, due to the following reasons. First, as demonstrated by McLeod et al. (2021), composite theoretical calibrations (e.g. D16, which combines $N2$ and [S II]/Hα) introduce non-negligible scatter compared to other empirical relation. Second, resolved metallicity maps are often susceptible to contamination from Diffuse Ionized Gas (DIG); Kumari et al. (2019) demonstrated that O3N2 is remarkably insensitive to DIG contamination (showing offsets of only $\sim$0.02 dex between H II and DIG regions), whereas indicators utilizing S II (such as D16 and Scal-PG16) are heavily affected, a finding we also confirm in Section 5.1. Third, both the O3N2-M13 and O3N2-C20 calibrations are derived from large IFS datasets (CALIFA and MaNGA, respectively), making them particularly well-suited for spatially-resolved studies like ours. Finally, O3N2-M13 is extensively employed in IFS studies investigating the rMZR (e.g. Sánchez et al. 2013; Barrera-Ballesteros et al. 2016; Erroz-Ferrer et al. 2019; Yao et al. 2022; Koller et al. 2024), ensuring our results are consistently comparable to the broader literature. However, we acknowledge the different results caused by different calibrations, and we ~~will~~ therefore discuss this effect in Section 5.1.

2.5 Local Moran-like Correlation Index (I_i)

To quantify how two spatially-resolved fields co-vary on local scales, we construct a "local Moran-like" product that combines the value of one property at a given position with the neighbourhood behaviour of a second property. This construction is inspired by the classical Moran's I statistic for spatial autocorrelation (Moran 1950) and its generalization: local indicators of spatial association (Anselin 1995). Such decomposing statistics were first introduced for astronomical use in Arnold & Wright (2024) to study the kinematic substructure of star clusters. Consider two generic maps $A(i)$ and $B(i)$ defined on the same set of valid spaxels i. We first compute the mean and standard deviation of each field over all valid locations, denoted (μ_A, σ_A) and (μ_B, σ_B), and form standardised residuals

$$\begin{cases} z_A(i) = \frac{A(i)-\mu_A}{\sigma_A}, \\ z_B(i) = \frac{B(i)-\mu_B}{\sigma_B}. \end{cases} \tag{11}$$

The offsets $z_A(i)$ and $z_B(i)$ measure the local departure of each spaxel from the global behaviour of the sample in units of the intrinsic scatter, after subtracting the azimuthally averaged radial profile (or the other secondary dependence) of each field. Positive values indicate that a given location lies above the mean in that field, while negative values indicate a deficit relative to the mean.

To embed this information in its spatial context, we next characterise the local environment of B around each spaxel i by averaging the z_B values of its neighbouring spaxels. In the present work, we adopt a simple eight-connected neighbourhood, so that the neighbour set $N(i)$ consists of the immediately adjacent spaxels in the two-dimensional grid and $N_i = 8$ for interior pixels. The local "spatial lag" of B at position i is then defined as

$$z_{{\rm lag},B}(i) \equiv \frac{1}{N_i} \sum_{j\in N(i)} z_B(j). \tag{12}$$

This quantity is a dimensionless measure of whether the environment surrounding i is systematically enhanced or suppressed in B relative to the global mean, again in units of the intrinsic dispersion: $z_{{\rm lag},B}(i) > 0$ implies that the neighbours tend to be above the mean in B, while $z_{{\rm lag},B}(i) < 0$ indicates a locally depressed environment.

The final local Moran-like product at each position is then defined as

$$I_i \equiv z_A(i)\, z_{{\rm lag},B}(i). \tag{13}$$

This scalar encapsulates both the sign and the strength of the local correspondence between the value of A at i and the surrounding behaviour of B. When $z_A(i)$ and $z_{{\rm lag},B}(i)$ share the same sign, $I_i > 0$ and the spaxel is "in agreement" with its environment: locations with $z_A(i) > 0$ and $z_{{\rm lag},B}(i) > 0$ correspond to regions that are elevated in A and embedded in an environment that is likewise elevated in B ("high–high"), whereas $z_A(i) < 0$ and $z_{{\rm lag},B}(i) < 0$ identify "low–low" regions that are suppressed with respect to both fields. Conversely, if $z_A(i)$ and $z_{{\rm lag},B}(i)$ have opposite signs, $I_i < 0$ and the spaxel is locally "inverted" with respect to its surroundings: a positive $z_A(i)$ combined with a negative $z_{{\rm lag},B}(i)$ signals a high value of A embedded in a depressed environment in B ("high–low"), while the opposite configuration corresponds to a "low–high" structure. The magnitude $|I_i|$ quantifies the statistical significance of this local pattern in units of the product of the two scatters. The resulting I_i map therefore provides a compact diagnostic of where, and how strongly, the field A is locally correlated or anti-correlated with the spatial environment of the field B, enabling a spatially resolved assessment of concordant versus discordant structures in any pair of astrophysical quantities. In this study, because A and B are measured on the Voronoi tessellation and then mapped back onto the native spaxel grid, the resulting I_i field is effectively smoothed on the bin scales but also suppressing structures and correlation signatures on scales smaller than a typical Voronoi bin size.

3 Results

3.1 The rMZR Relation

Before analysing the ensemble of resolved relations derived from combining all MAUVE–MUSE galaxies, we note that one of the 14 target galaxies, NGC 4383, is identified as a metallicity outlier relative to the rest of the sample. Across all four strong-line prescriptions adopted in this work (D16, Scal-PG16, O3N2-M13, and O3N2-C20), the oxygen abundances of star-forming spaxels in NGC 4383, especially the high-$\Sigma_{\rm SFR}$ spaxels (star-forming disc), are systematically offset low by at least 0.2 dex compared to the locus defined by the other 13 galaxies. This behaviour persists under our standard **HII** masking and across calibrators, and is therefore unlikely to be driven by methodological systematics. Because this single galaxy would otherwise dominate the low-metallicity tail and bias the ensemble trends, NGC 4383 is likely not a representative of the "typical" disk population in our sample. We therefore define a fiducial ensemble sample that excludes NGC 4383 for the purposes of fitting

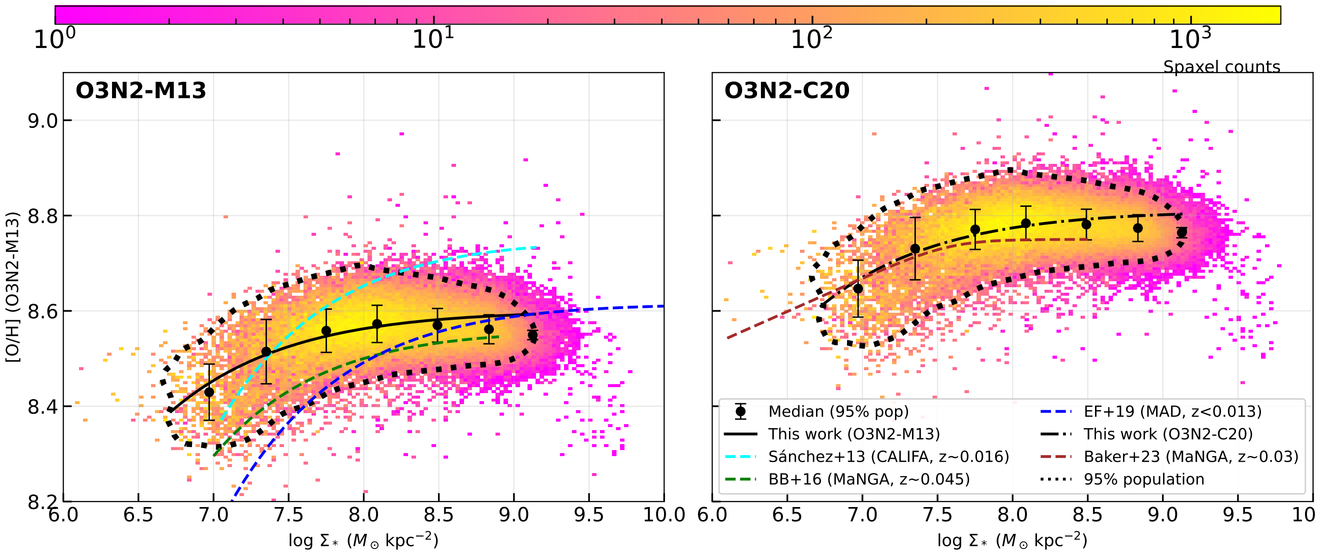

Figure 2. The resolved mass–metallicity relation (rMZR) for 13 MAUVE–MUSE galaxies (excluding NGC 4383). The left panel shows the rMZR derived using the O3N2-M13 calibration, while the right panel shows the rMZR using the O3N2-C20 calibration. The color scale indicates the number of spaxel. The dotted contours indicate the 95% population bounds, and the black points with error bars show the median metallicity within the contours. The solid and dot-dashed black curves represent the best-fitting rMZR to our data using Equation 14 under O3N2-M13 and O3N2-C20 prescriptions, respectively. For comparison, we show literature relations: the blue dashed curve from MAD data ($z < 0.013$; Erroz-Ferrer et al. 2019), the cyan dashed curve from CALIFA data ($z \sim 0.016$; Sánchez et al. 2013), the green dashed curve from MaNGA data ($z \sim 0.045$; Barrera-Ballesteros et al. 2016) in the left panel, and the brown dashed curve from MaNGA data ($z \sim 0.03$; Baker et al. 2023) in the right panel. Note that Baker et al. (2023) adopted the "combined" prescription from Curti et al. (2020).

Calibration	a	b	c
O3N2-M13	8.597 ± 0.001	0.0024 ± 0.00018	10.00 ± 0.057
O3N2-C20	8.807 ± 0.001	0.0023 ± 0.00017	10.00 ± 0.056

Table 2. Best-fitting parameters of the resolved MZR functional form in Equation 14 for the O3N2-M13 and O3N2-C20 metallicity prescriptions.

and benchmarking the ensemble trends. Hereafter, when we refer to '13 galaxies' or the 'ensemble sample', NGC 4383 is excluded unless explicitly stated otherwise. We ~~will~~ return to discuss its properties in Section 5.3.

After combining all **HII**-selected spatial measurements from the 13 galaxies into a single ensemble, the resulting spaxel population follows the familiar rMZR shape reported in previous IFS surveys (e.g. Sánchez et al. 2013; Barrera-Ballesteros et al. 2016; Sánchez et al. 2019; Erroz-Ferrer et al. 2019; Yao et al. 2022; Baker et al. 2023; Koller et al. 2024): $Z_{\rm gas}$ increases steeply with Σ_* at the low-mass end and gradually transitions to a shallower, saturation-like behaviour toward high Σ_*. In Figure 2, we present this locus using the 95% population contour and its median trend (black points), shown for both the O3N2-M13 (left) and O3N2-C20 (right) calibrations. Within the 95% contour, we perform a nonlinear least-squares fit by adopting the functional form from Sánchez et al. (2013) to describe the trend of rMZR:

$$[{\rm O/H}] = a + b\,(\log_{10}\Sigma_* - c)\,e^{-(\log_{10}\Sigma_* - c)}, \quad (14)$$

The best-fitting parameters for our 13 MAUVE–MUSE galaxies are listed in Table 2.

In the left panel of Figure 2, we compare our best-fitting rMZR obtained with the O3N2-M13 calibration to three other spatialy-resolved local-universe surveys that adopt the same metallicity prescription. Overall, the MAUVE–MUSE relation shows the familiar saturating rMZR behaviour but exhibits the shallowest low-Σ_* rise among the plotted studies. Relative to the MAD relation (dark blue dashed curve) of Erroz-Ferrer et al. (2019) at comparable (~100pc) resolution, we recover a similar high-Σ_* plateau at $12+\log_{10}({\rm O/H}) \simeq 8.6$, whereas the MAD curve declines more steeply toward low Σ_*, reaching ~0.3 dex lower metallicity around $\log_{10}(\Sigma_*/M_\odot\,{\rm kpc}^{-2}) \simeq 7.0$. Compared to the MaNGA-based relation (green dashed curve) of Barrera-Ballesteros et al. (2016), our rMZR is systematically higher by ~0.05–0.1 dex over the entire dynamic range of $\log_{10}(\Sigma_*)$. The comparison to CALIFA relation (cyan dashed curve) by Sánchez et al. (2013) is non-monotonic: in the intermediate- and high-mass regimes ($7.5 \lesssim \log_{10}(\Sigma_*/M_\odot\,{\rm kpc}^{-2}) \lesssim 9.0$) our curve lies below Sánchez et al. (2013), but at $\log_{10}(\Sigma_*/M_\odot\,{\rm kpc}^{-2}) \lesssim 7.5$ it becomes higher, so that our rMZR is consistent with Barrera-Ballesteros et al. (2016) and implies a comparatively shallow low-Σ_* metallicity decline in MAUVE–MUSE data. Part of this offset may reflect differences in sample selection: the early CALIFA analysis of Sánchez et al. (2013) is effectively complete only above $\log_{10}(M_*/M_\odot) \gtrsim 9.5$. As a result, its high-$\Sigma_*$ component is weighted toward more massive, more metal-rich discs, which could elevate the CALIFA normalization relative to our late-type Virgo spirals in MAUVE.

The right panel shows that adopting O3N2-C20 shifts the best-fitting rMZR upward by ~0.2 dex relative to O3N2-M13 at fixed Σ_*, underscoring that the absolute normalisation is calibration-dependent (see further discussion in Section 5.1). In this representation, the "combined" prescription used by Baker et al. (2023) yields a best-fitting relation (brown dashed curve) that is highly consistent with our O3N2-C20 fit, particularly at $\log_{10}(\Sigma_*/M_\odot\,{\rm kpc}^{-2}) \lesssim 7.5$. Aditionally, although not shown here, our O3N2-M13 relation has a similar overall curvature to other IFS measurements at larger distances and/or intermediate redshifts (e.g. Yao et al. 2022; Koller et al. 2024) but with a higher normalisation; as discussed by Koller et al. (2024), such offsets are unlikely to be explained by redshift evolution of the rMZR alone. Finally, as noted by Koller et al. (2024), we acknowledge that differences in Σ_* estimation (i.e. SSP assumptions) and related pipeline choices can also contribute to survey-to-survey rMZR discrepancies.

3.2 rMZR's Secondary Dependence of rMZR on $\Sigma_{\rm SFR}$

Colour-coding the rMZR by $\Sigma_{\rm SFR}$ reveals a clear secondary dependence on local star formation intensity (see the left column of Figure 3). At fixed Σ_*, high-$\Sigma_{\rm SFR}$ spaxels tend to lie at higher $Z_{\rm gas}$ in the high-Σ_* regime, whereas the ordering reverses at low Σ_*, where lower-$\Sigma_{\rm SFR}$ spaxels are preferentially more metal-rich. The transition between these regimes occurs at $\log_{10}(\Sigma_*/M_\odot\,{\rm kpc}^{-2}) \simeq 7.5$–8.0. While the exact reversal point is imprecise due to limited statistics in the narrow overlap region, its location is consistent with the inversion reported by MAGPI by Koller et al. (2024), who found a clear sign change near $\log_{10}(\Sigma_*) \sim 7.8$ using O3N2-M13 prescription as well. The MAUVE–MUSE data therefore support a mass-dependent inversion in the resolved $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ coupling, suggesting that the dominant driver of the secondary dependence (the balance between metal enrichment and dilution) may differ between low- and high-Σ_* environments. Similarly, although not a strong signal, under the prescriptions of Curti et al. (2020), MaNGA-based analysis of Baker et al. (2023) also appears consistent with a flipped correlation in the spatially-resolved scenario, with the transition occurring at $\log_{10}(\Sigma_*/M_\odot\,{\rm kpc}^{-2}) \approx 7.5$–7.7 (their Figure 2), in good agreement with our inferred range. Furthermore, adopting both D16 and Scal-PG16 calibrations, WL21 reported a predominantly positive local $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ correlation using the MUSE Atlas of Disks (MAD) data at $\sim$ 100 pc resolution, whereas with MaNGA data on global galactic scales they found an overall negative correlation.

To visualise metallicity's dependence on $\Sigma_{\rm SFR}$ more directly, the right column of Figure 3 shows $Z_{\rm gas}$ as a function of $\Sigma_{\rm SFR}$ in bins of Σ_*. For each Σ_* bin we also compute the Spearman rank correlation coefficient ρ between $Z_{\rm gas}$ and $\Sigma_{\rm SFR}$ after removing the 3σ outliers. The lowest-Σ_* bins ($\log_{10}\Sigma_*$ at 6.5 - 7.0) exhibit the strongest anti-correlation at Spearman coefficient $\rho = -0.41$, with metallicity decreasing as $\Sigma_{\rm SFR}$ rises. With increasing Σ_*, the median trends flatten progressively, and ultimately the relation becomes positively correlated, such that regions of elevated star formation are also metal-enhanced. This systematic, mass-ordered sequence from negative through flat to positive slopes encapsulates the rMZR's second dependence on $\Sigma_{\rm SFR}$ and motivates the offset and spatial-correlation analyses undertaken in the next section.

3.3 Offset Relation and Local Moran-like Correlation Map

To isolate the secondary coupling between metallicity and star formation from their primary dependence on stellar mass surface density, we construct mass-normalised offset maps for each galaxy. Specifically, within each galaxy we divide all valid **HII** spaxels into 12 equal-width bins in $\log_{10}\Sigma_*$ (illustrated for one of MAUVE–MUSE targets, NGC 4192, in the bottom-right panel of Figure 4). For every bin m, we compute the mean spaxel values $\langle \log_{10}\Sigma_{\rm SFR}\rangle_m$ and $\langle [{\rm O/H}]_{\rm O3N2-M13}\rangle_m$, and define spaxel-level offsets

$$\begin{cases} \Delta\log_{10}\Sigma_{\rm SFR}(i) \equiv \log_{10}\Sigma_{\rm SFR}(i) - \langle \log_{10}\Sigma_{\rm SFR}\rangle_m, \\ \Delta[{\rm O/H}](i) \equiv [{\rm O/H}](i) - \langle [{\rm O/H}]\rangle_m, \end{cases} \tag{15}$$

where spaxel i belongs to bin m. By construction, these offsets remove the mean rMZR and resolve star forming main sequence (rSFMS) trends within each Σ_* interval, leaving only the relative fluctuations of $\Sigma_{\rm SFR}$ and $Z_{\rm gas}$ at fixed local stellar mass density. The resulting maps therefore trace spatial variations that are not driven by the underlying Σ_* distribution and its associated large-scale gradients (e.g. see top panels of Figure 4).

Using these mass-normalised quantities, we also measure the residual $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ coupling by computing, in each Σ_* bin of each galaxy, the Spearman rank correlation coefficient between $\Delta[{\rm O/H}]_{\rm O3N2-M13}$ and $\Delta\log_{10}\Sigma_{\rm SFR}$ after removing the 3σ outliers. The resulting $\rho(\Sigma_*)$ trends are shown in Figure 5; each reported ρ has corresponding p-value $\lesssim 0.01$, implying statistically significant correlation between the two quantities. For the ensemble sample (black curve), ρ increases smoothly from negative values at low Σ_* to positive values at high Σ_*: rising from $\rho \approx -0.35$ at $\log_{10}\Sigma_* \sim 7$ to $\rho \approx +0.3$ at $\log_{10}\Sigma_* \gtrsim 8$, with a zero-crossing within $\log_{10}(\Sigma_*/M_\odot\,{\rm kpc}^{-2}) \simeq 7.5$–8.0. Individual galaxies follow the same qualitative ordering: systems that predominantly sample the low-Σ_* regime (e.g. NGC 4396, NGC 4457, and part of NGC 4192) contribute to the anti-correlated branch, whereas galaxies whose **HII** spaxels lie mainly at high Σ_* populate the positively correlated regime. Notably, NGC 4192 spans a broad Σ_* range that straddles the inversion point and exhibits a clear internal sign reversal, indicating that the transition is intrinsic to the subtle physics within galaxies rather than an artefact of combining different galaxy populations.

We further assess the spatial locality of this residual coupling by constructing the local bivariate Moran-like product I_i defined in Section 2.5. As an illustrative example, the bottom-left panel of Figure 4 shows the I_i map for NGC 4192, derived from the corresponding $\Delta\log_{10}\Sigma_{\rm SFR}$ and $\Delta[{\rm O/H}]$ fields. As indicated by Equation 15, these two fields are defined as residuals at fixed Σ_*, i.e. after subtracting, in each Σ_* bin, the trends of mean rMZR and rSFMS. In practice, this removes the dominant radial (mass-driven) profiles, so that the I_i map traces purely local spatial correspondence between star formation and metallicity fluctuations. By construction, I_i is the product of two standardised quantities, $z_{\Delta\log_{10}\Sigma_{\rm SFR}}(i)$ and $z_{{\rm lag},\Delta[{\rm O/H}]}(i)$, each expressed in units of each Σ_* bin's standard deviation of the corresponding field. Thus $|I_i| \sim 1$ corresponds to the joint product of two $\sim 1\sigma$ deviations (one at the spaxel position, one in its local environment), whereas $|I_i| \ll 1$ indicates that at least one of the two is close to the global mean. Therefore, while the choice is inherently arbitrary, we adopt $|I_i| = 1$ as a convenient, dimensionless threshold to distinguish weak or typical patterns from stronger local correspondence.

For NGC 4192, most spaxels in Figure 4 have $|I_i| \leq 1$ (grey), implying weak or ambiguous local correspondence, while a non-negligible fraction ($\sim$ 20%) show significant local correlation ($I_i > 1$, red: higher (lower) $\Sigma_{\rm SFR}$ with higher (lower) $Z_{\rm gas}$) or anti-correlation ($I_i < -1$, blue: higher (lower) $\Sigma_{\rm SFR}$ with lower (higher) $Z_{\rm gas}$). These two tails are of comparable magnitude, with 7.5% of spaxels ($N = 11002$) having $I_i > 1$ and 11.8% ($N = 17221$) having $I_i < -1$, while the majority (80.7%; $N = 117933$) remain in the weak/ambiguous regime. This near-balance between correlated and anti-correlated H II spaxels is consistent with NGC 4192 being the clearest "inversion" case in our sample: depending on Σ_*, the net $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ coupling can change sign in Figure 5. Interpreting high-$\Sigma_{\rm SFR}$ peaks as tracing the cores of individual H II regions, the coexistence of red and blue structures within the same Σ_* bins suggests that the local $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ coupling is spatially heterogeneous: once mass trends of these two properties are removed, neighbouring regions of comparable Σ_* can display either concordant or discordant metallicity responses to star formation. Moreover, the highest $|I_i|$ values are spatially coincident with bright Hα/high-$\Sigma_{\rm SFR}$ clumps that satisfy our **HII** BPT selection, whereas diffuse low-$\Sigma_{\rm SFR}$ regions tend to show $|I_i| \lesssim 1$, supporting that the observed signal is associated with genuine H II regions (or clumps of them) rather than being driven by

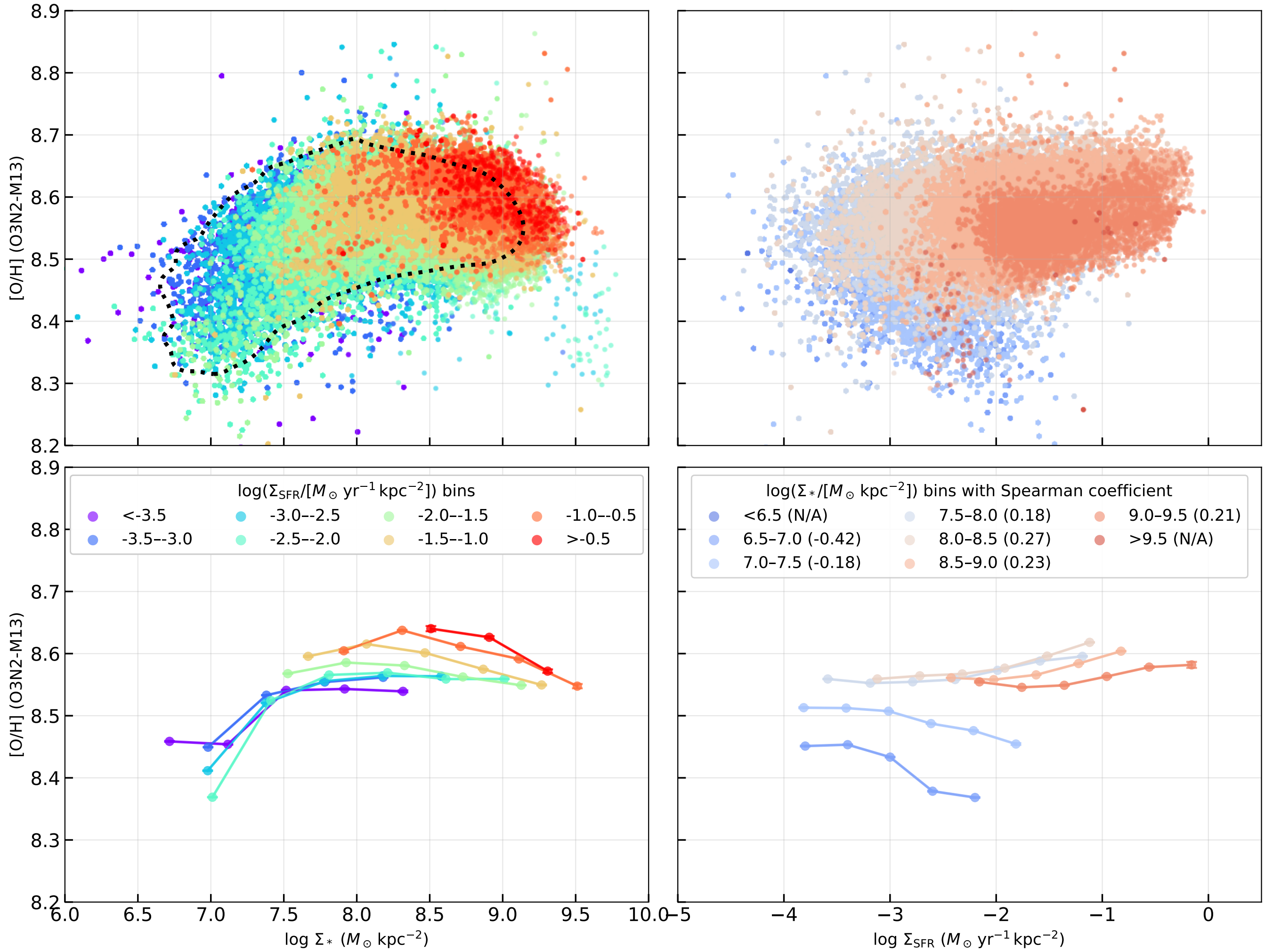

Figure 3. The left column shows the rMZR colour-coded by $\Sigma_{\rm SFR}$ for 13 MAUVE–MUSE galaxies (excluding NGC 4383), while the right column shows the $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ relation colour-coded by Σ_*. The upper panels display the scatter plots, while the lower panels show the corresponding median trends of each scatter plot. Similar to Figure 2, we use the black contour to enclose the 95% population of **HII** spaxels. Within this contour, we present the median trends with errorbars indicating the stardard error of the median (SEM; typical value of $\lesssim 0.01$). Note that due to insufficient statistics, the median trends of the lowest and highest Σ_* bins in the bottom right panel are not shown.

DIG contamination. These maps show that (i) the sign of the $\Delta Z_{\rm gas}$–$\Delta\Sigma_{\rm SFR}$ coupling is set by heterogeneous local physical processes, so correlated and anti-correlated H II regions can coexist at fixed Σ_*, even in neighboring regions; and (ii) the observed mass-dependent behaviour arises only as an emergent population trend when one aggregates enough spaxels to sample both sides of the transition around $\log_{10}(\Sigma_*/M_\odot\,{\rm kpc}^{-2}) \sim 7.5$–$8.0$.

4 The Origin of $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ (Anti-)Correlation in a Gas-regulator Framework

To interpret the varying sign of the $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ relation and its residual $\Delta Z_{\rm gas}$–$\Delta\Sigma_{\rm SFR}$ coupling, we adopt a spatially-resolved gas–regulator ("bathtub") framework formulated entirely in spaxel-by-spaxel variations. Throughout this section, all quantities are written as surface densities to reflect the fact that our observational constraints are local (~ 100 pc resolution) rather than global. The model is intended to describe the evolution of the gas regulator in the local universe, over timescales that are short compared to the characteristic orbital period in the disc ($\sim 200-300$ Myr). In practice, the variations that matter for our observed $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ (anti-)correlation are closer to the lifetimes of individual H II regions and local dynamical or mixing timescales (i.e. a few tens of Myr), as suggested by the fact that the strongest signal (strongest correlation between [O/H] and $\Sigma_{\rm SFR}$) is associated with bright Hα/high-$\Sigma_{\rm SFR}$ clumps. Over such intervals, the accumulated growth of the stellar mass surface density Σ_* is negligible compared to its present-day value (or at most comparable to observational uncertainties $\lesssim 0.05$ dex), so we neglect the time evolution of Σ_* and treat it as a fixed control parameter. We therefore do not attempt to reconstruct the full past star formation or metal enrichment history of each region. Instead, we assume that within each galaxy, and at fixed Σ_*, different H II spaxels within a galaxy began their recent evolution from broadly similar initial conditions and have since evolved under the same set of governing physical conditions.

Within this framework, the time variable, t, plays the role of an evolution parameter that controls the instantaneous state of the local regulator (gas mass, SFR, metallicity, gas supply properties) at a given location. Mathematically, even though we observe different

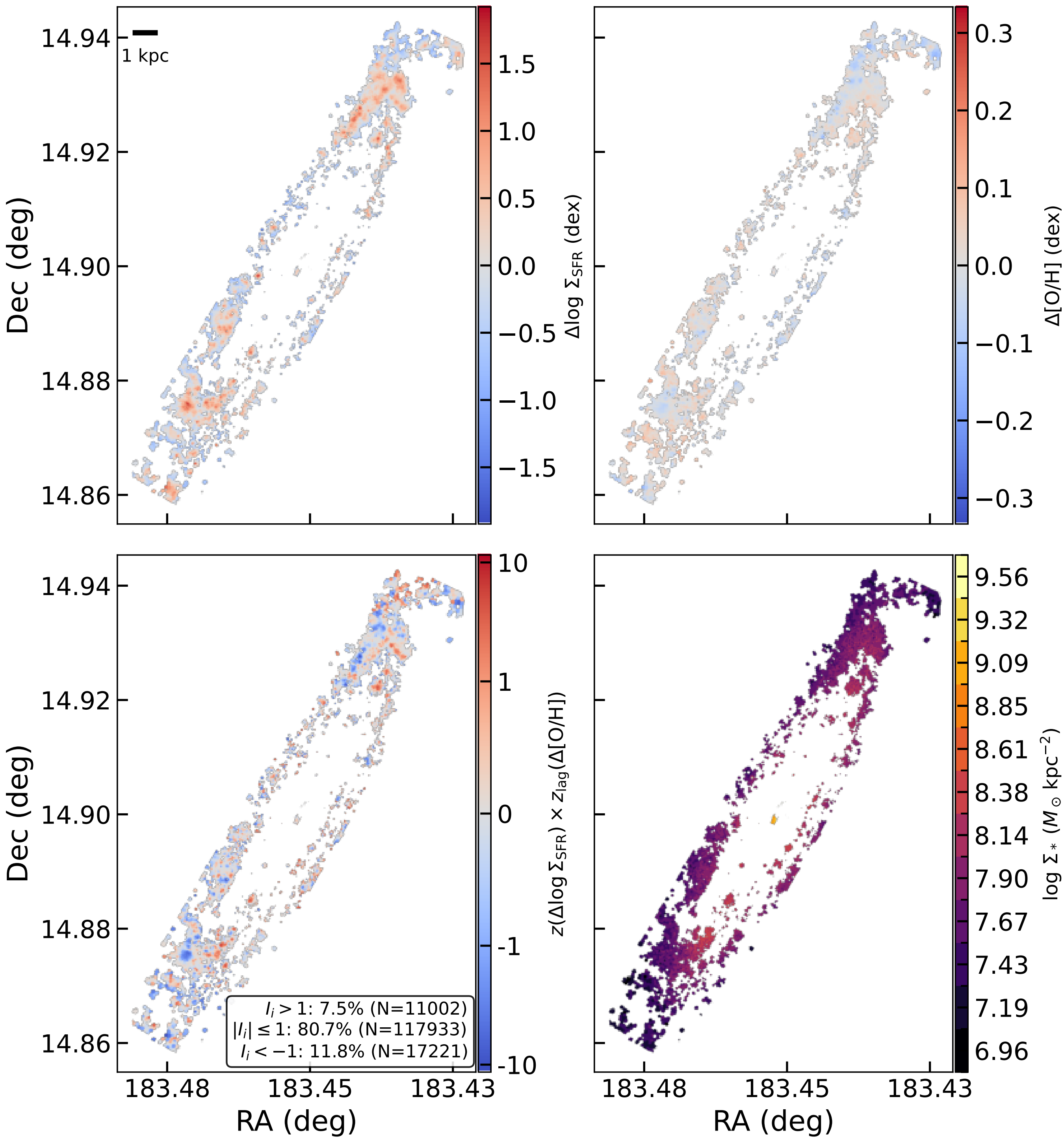

Figure 4. Example offset and local Moran-like correlation maps for **HII** spaxels of NGC 4192. The top panels show the offsets $\Delta \log_{10} \Sigma_{\rm SFR}$ (left) and $\Delta[{\rm O/H}]_{\rm O3N2-M13}$ (right) after removing the mean trends with Σ_* (see text). The bottom-left panel presents the local bivariate Moran-like product I_i constructed from these two offset fields, highlighting spatially local correlation (red) and anti-correlation (blue). The bottom-right panel shows $\log_{10} \Sigma_*$ for the **HII** spaxels used in the other panels, divided into 12 equal-width bins.

spaxels at the same cosmic time, each with its own set of properties, P (here $P \in \{\Sigma_*, \Sigma_{\rm SFR}, Z_{\rm gas}, {\rm etc.}\}$), it is useful to think of these spatial variations as sampling different points along a common family of time-dependent solutions. Equivalently, given any pair of spaxels within the same Σ_* bin, the differences in their local properties can be interpreted as the result of having evolved for different effective "phases" or having experienced different recent perturbations under the same functional form of the regulator equations. In this sense, the time coordinate, t, may be regarded as a parameter that indexes the instantaneous value of each property and allows us to trace how fluctuations in one property, P, induce corresponding responses in another. What follows is therefore a local, time-dependent gas-regulator

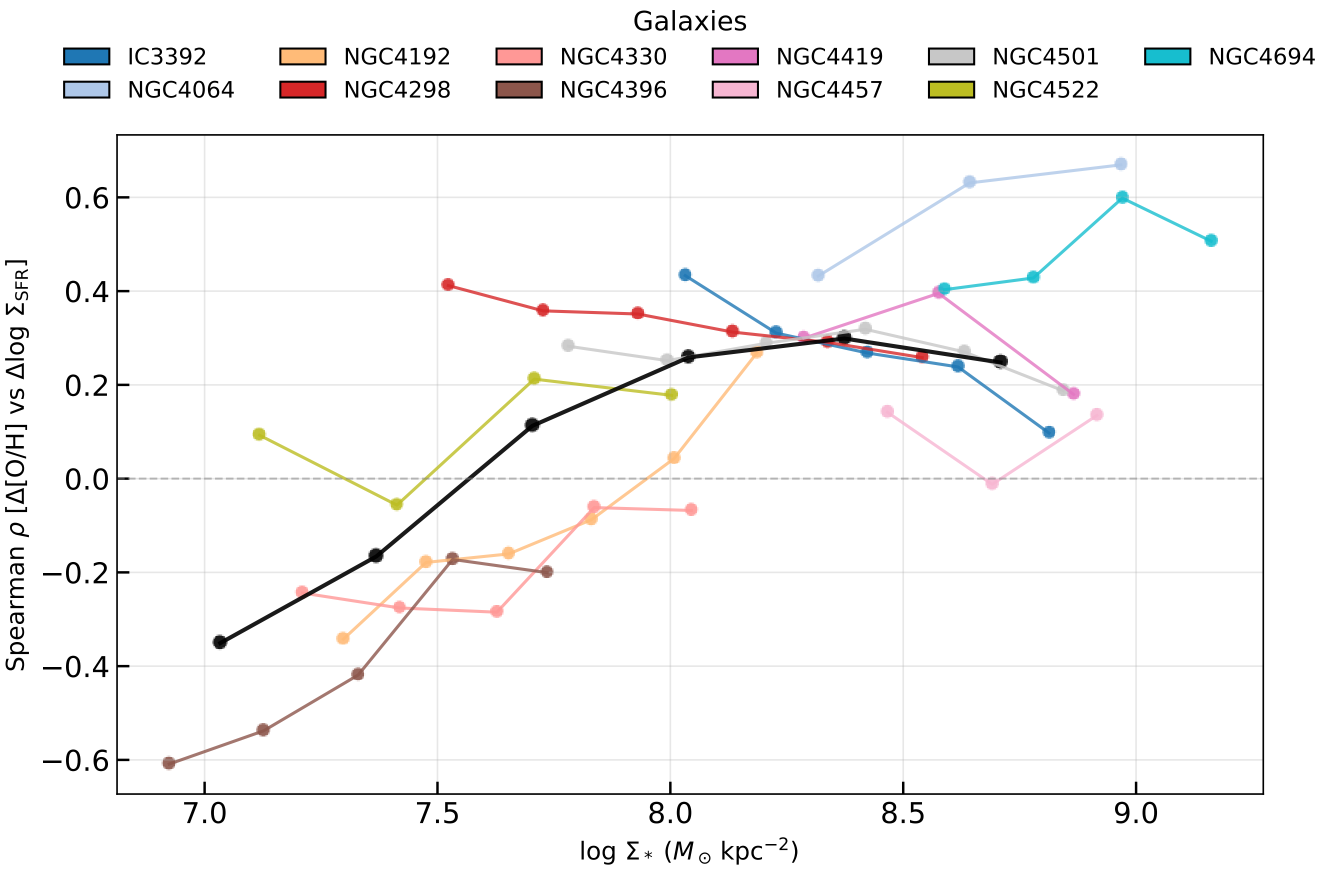

Figure 5. Spearman rank correlation coefficient ρ between $\Delta[\mathrm{O/H}]_{\rm O3N2-M13}$ and $\Delta \log_{10}\Sigma_{\rm SFR}$ as a function of $\log_{10}\Sigma_*$, measured in discrete Σ_* bins for 13 MAUVE–MUSE galaxies (excluding NGC 4383). Each coloured curve corresponds to one galaxy; the Spearman trends of NGC 4293 and NGC 4698 are not shown here due to insufficient statistics. The black curve shows the ensemble result obtained by combining all **HII** spaxels from the 13 galaxies.

model for the coupled variations of $\Sigma_{\rm gas}$, $\Sigma_{\rm SFR}$, and $Z_{\rm gas}$ at fixed Σ_*, which we use as a physical language for organizing the spatially-resolved fluctuations measured in the MAUVE–MUSE data.

We express the star-formation rate surface density as

$$\Sigma_{\rm SFR}(t) \equiv \frac{\Sigma_{\rm g}(t)}{\tau_{\rm dep}(t)}, \tag{16}$$

where $\Sigma_{\rm g}(t)$ is the gas surface density and $\tau_{\rm dep}(t)$ the local depletion time. Then, we use the following two differential equations to describe our spatially-resolved framework. First, the gas mass surface density obeys

$$\frac{d\Sigma_{\rm g}(t)}{dt} = \Sigma_\Phi(t) - (1-R)\Sigma_{\rm SFR}(t) - \Sigma_{\rm out}(t), \tag{17}$$

where $\Sigma_\Phi(t)$ is the *gas supply rate (GSR) surface density*§, i.e. the rate at which the region gains fresh gas per unit area, R is the fraction of the stellar mass which is immediately returned by massive stars, and $\Sigma_{\rm out}(t)$ is the outflow rate surface density which accounts for gas removal driven by both internal feedbacks and environmental mechanisms like stripping.

Then, the surface density of metals (Σ_Z) follows

$$\frac{d\Sigma_Z(t)}{dt} = y(1-R)\,\Sigma_{\rm SFR}(t) - \big[(1-R)\Sigma_{\rm SFR}(t) + \Sigma_{\rm out}(t)\big]Z(t) + \Sigma_\Phi(t)\,Z_\Phi, \tag{18}$$

with metal yield per stellar mass, y, and metallicity of the supplied gas, Z_Φ (i.e., background metallicity).

Defining the gas-phase metallicity as

$$Z(t) \equiv \frac{\Sigma_Z(t)}{\Sigma_{\rm g}(t)}, \tag{19}$$

we obtain

$$\frac{dZ(t)}{dt} = \frac{1}{\Sigma_{\rm g}(t)}\left[y(1-R)\,\Sigma_{\rm SFR}(t) + \big(Z_\Phi - Z(t)\big)\,\Sigma_\Phi(t)\right], \tag{20}$$

so that at any location, the metallicity responds to enrichment from ongoing star formation and dilution by the supplied gas.

It is convenient to introduce the gas supply timescale

$$\tau_\Phi(t) \equiv \frac{\Sigma_{\rm g}(t)}{\Sigma_\Phi(t)} \;\leq\; \tau_{\rm dep}(t), \tag{21}$$

which measures how quickly the local gas reservoir would be replen-

§ We avoid the term "inflow rate surface density" because inflow usually denotes pristine, galaxy-scale accretion from the circumgalactic medium (CGM), whereas on local scales it is more appropriate to describe a region's or giant molecular cloud (GMC)'s ability to gain gas supply from its ambient ISM; moreover, for our observed Virgo members, the global inflow is likely suppressed or truncated (starvation; Larson et al. 1980).

Symbol	Meaning
t	Time / evolution parameter indexing the instantaneous state of the local regulator (used to organise spaxel-to-spaxel fluctuations).
Σ_*	Stellar mass surface density; treated as a fixed control parameter within a given Σ_* bin over the short timescales of interest.
$\Sigma_g(t)$	Gas mass surface density (the local gas reservoir).
$\Sigma_{SFR}(t)$	Star formation rate (SFR) surface density.
$\tau_{dep}(t)$	Gas depletion time, defined by $\Sigma_{SFR} \equiv \Sigma_g/\tau_{dep}$.
$\Sigma_\Phi(t)$	Gas supply rate (GSR) surface density: gas accretion rate at which the region gains gas per unit area from its surroundings (local supply/replenishment).
$\tau_\Phi(t)$	Gas supply timescale, $\tau_\Phi \equiv \Sigma_g/\Sigma_\Phi$ (replenishment time of the local gas reservoir at the current GSR).
$\Sigma_{out}(t)$	Outflow rate surface density (gas removed by feedback and/or environmental processes such as stripping).
R	Return fraction: fraction of newly formed stellar mass promptly recycled back to the ISM.
$\Sigma_Z(t)$	Metal surface density in the gas phase (in the same metal species traced by Z; here effectively oxygen).
$Z(t)$	Gas-phase metallicity of the local reservoir, $Z \equiv \Sigma_Z/\Sigma_g$ (model variable corresponding to an oxygen abundance).
y	Metal yield per unit stellar mass formed.
Z_Φ	Metallicity of the supplied gas (background/pre-enrichment level of the gas entering the region).
$Z^\dagger(t)$	Instantaneous target metallicity (driver/attractor for Z), $Z^\dagger \equiv Z_\Phi + y(1-R)\,\Sigma_{SFR}/\Sigma_\Phi$.

Table 3. Symbols used in the spatially-resolved gas-regulator model.

ished by the current GSR. Using Equation 16 and Equation 21, Equation 20 yields

$$\frac{dZ(t)}{dt} = \frac{1}{\tau_\Phi(t)}\left[Z_\Phi + y(1-R)\frac{\Sigma_{SFR}(t)}{\Sigma_\Phi(t)} - Z(t)\right]. \quad (22)$$

This motivates the definition of the *instantaneous target metallicity*

$$Z^\dagger(t) \equiv Z_\Phi + y(1-R)\frac{\Sigma_{SFR}(t)}{\Sigma_\Phi(t)}, \quad (23)$$

so that Equation 22 becomes

$$\frac{dZ(t)}{dt} = \frac{1}{\tau_\Phi(t)}\left[Z^\dagger(t) - Z(t)\right] \iff Z^\dagger(t) = Z(t) + \frac{dZ(t)}{dt}\tau_\Phi(t). \quad (24)$$

This explicitly shows that $Z^\dagger(t)$ acts as the *driver* of the metallicity evolution compared to the real-time metallicity $Z(t)$ (the observed metallicity): $Z^\dagger(t)$ is the metallicity that the system would reach if the current rate of change dZ/dt were sustained over one gas supply timescale. Only when $Z^\dagger(t) = Z(t)$ does Equation 24 imply $dZ/dt = 0$, corresponding to a *local minimum or maximum* of $Z(t)$ rather than a long-lived equilibrium.

With these definitions, the observed spaxels can be separated conceptually into two complementary cases: either reaching (Section 4.1) or evolving toward/away from (Section 4.2) the local extremum stages. Table 3 summarises the notation used in this patially-resolved gas-regulator model.

4.1 Case I: Correlation at $Z = Z^\dagger$

For spaxels observed at a series of local extremum times $t = t_m$, the real-time metallicity satisfies $dZ/dt|_{t_m} = 0$, hence $Z(t_m) = Z^\dagger(t_m)$, with

$$Z^\dagger(t_m) = Z_\Phi + y(1-R)\frac{\Sigma_{SFR}(t_m)}{\Sigma_\Phi(t_m)}. \quad (25)$$

Taking a first-order expansion of Equation 25 yields

$$\delta Z^\dagger(t_m) = \left[Z^\dagger(t_m) - Z_\Phi\right]\left[\delta\log_{10}\Sigma_{SFR}(t_m) - \delta\log_{10}\Sigma_\Phi(t_m)\right], \quad (26)$$

so the sign of the metallicity fluctuation at extrema is controlled by the competition between spatial variations in SFR and GSR. If Σ_{SFR} varies more strongly than Σ_Φ across neighbouring spaxels, then $\delta Z^\dagger$ follows $\delta\log_{10}\Sigma_{SFR}$ and a positive Z_{gas}–Σ_{SFR} correlation is expected. If instead GSR variations dominate while Σ_{SFR} and Σ_Φ co-vary, dilution drives $\delta Z^\dagger$ opposite to $\delta\log_{10}\Sigma_{SFR}$, yielding an anti-correlation. Where the two contributions are comparable, $\delta Z^\dagger$ is small and little correlation is produced.

Using Equation 16 and Equation 21, Equation 26 is equivalently

$$\delta Z^\dagger(t_m) \propto \delta\log_{10}\tau_\Phi(t_m) - \delta\log_{10}\tau_{dep}(t_m), \quad (27)$$

highlighting that, at metallicity extrema, the correlation sign is set by whether variations in the gas supply timescale outpace those in depletion time. Furthermore, if we generalize (i.e., substitute) t_m with t, which in fact describes the change of instantaneous target metallicity rather than the real-time metallicity at extrema, we still have the same functional forms of Equation 26 and Equation 27. This tells us that the change of instantaneous target metallicity is also determined by the comparison of the relative change magnitude between the local SFR and GSR. We will return to this relation later in our discussion.

4.2 Case II: Correlation at $Z \neq Z^\dagger$

In reality, most spaxels are not observed exactly at a local metallicity extremum. To relate their real-time metallicity to the driver $Z^\dagger$, we expand around the nearest extremum time t_m by rewriting $t = t_m + \zeta$ (ζ is an arbitrary time interval away from the extremum time t_m) and, for any quantity P ($P \in \{\Sigma_{SFR}, Z, Z^\dagger, \text{etc.}\}$), defining

$$\Delta_P(\zeta) \equiv P(t_m + \zeta) - P(t_m). \quad (28)$$

Linearising Equation 24 about t_m yields a first-order response,

$$\frac{d\,\Delta_Z(\zeta)}{d\zeta} + \frac{\Delta_Z(\zeta)}{\tau_\Phi(t_m)} = \frac{\Delta_{Z^\dagger}(\zeta)}{\tau_\Phi(t_m)}, \quad (29)$$

whose solution is

$$\Delta_Z(\zeta) = \frac{1}{\tau_\Phi(t_m)}\int_0^\zeta \exp\left[-\frac{\zeta-\nu}{\tau_\Phi(t_m)}\right]\Delta_{Z^\dagger}(\nu)\,d\nu. \quad (30)$$

Equation 30 shows that $\Delta_Z(\zeta)$ is a low-pass-filtered version of the driver $\Delta_{Z^\dagger}(\zeta)$: the real-time metallicity responds as an exponential moving average of the target metallicity, smoothed over the gas supply timescale $\tau_\Phi(t_m)$. Physically, this "memory" arises because $Z(t)$ cannot respond instantaneously to changes in $Z^\dagger(t)$, due to finite delays in metal injection from recent star formation and in turbulent mixing (metal diffusion) through the ISM. In practice, because of detection procedures and measurement uncertainties, our observations predominantly probe the regime $\zeta \gtrsim \tau_\Phi(t_m)$, i.e. the time-averaged response rather than an instantaneous snapshot of $Z^\dagger$.

For such observed spaxels, $\delta Z(t)$ follows the recent behaviour of $\delta Z^\dagger(t)$ and therefore inherits the same sign set by star formation versus gas supply variability. Inserting Equation 30 into Equation 24, yields the consistent prediction as Equation 26 and Equation 27:

$$\begin{aligned}\delta Z(t) &\propto \delta\log_{10}\big(\Sigma_{SFR}(t)\big) - \delta\log_{10}\big(\Sigma_\Phi(t)\big)\\ &\propto \delta\log_{10}\big(\tau_\Phi(t)\big) - \delta\log_{10}\big(\tau_{dep}(t)\big).\end{aligned} \quad (31)$$

Thus, away from extrema, the observed local $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ (anti-)correlation is still governed by whether spatial variations are driven primarily by changes in star-formation efficiency (through $\tau_{\rm dep}$) or by changes in the gas supply rate (through τ_Φ), with the finite response time encoded in τ_Φ introducing additional scatter and spatially varying phase lags.

4.3 Interpretation of the Observed (Anti-)Correlation

Within this framework, our resolved MAUVE–MUSE trends in Figure 5 can be interpreted as the outcome of a competition between GSR-driven and SFR-driven variability of each gas regulator. Observationally, in high-Σ_* regions (typically inner discs), dense gas and short depletion times make spatial variations in $\Sigma_{\rm SFR}$ or $\tau_{\rm dep}$ relatively strong, while the ability to gain fresh gas (i.e., the GSR) is comparatively steady from spaxel to spaxel. In this regime, Equation 31 favours a positive $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ correlation: fluctuations in SFR dominate over variations in GSR, and oxygen abundance increases with $\Sigma_{\rm SFR}$ at fixed Σ_*. On the other hand, at low Σ_* (outer discs), especially in the Virgo environment where the local gas supply can fluctuate strongly due to environmental processes, spatial variations in τ_Φ become more important. When GSR fluctuations dominate while $\Sigma_{\rm SFR}$ and Σ_Φ co-vary, the dilution associated with enhanced supply changes $Z_{\rm gas}$ to lower values in regions of relatively higher SFR, driving the correlation toward zero or negative values. In between these extremes, where variations in τ_Φ and $\tau_{\rm dep}$ are comparable, the net effect on metallicity is small and the $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ relation becomes weak or flat.

WL21 also explain this (anti-)correlation from a theoretical point of view. Their model attributes the positive case (in ~ 100 pc scales) to time-variable SFE acting against a relatively steady gas inflow rate, and the negative case (at galactic scales) to strongly time-variable inflow acting against nearly constant SFE. Our MAUVE–MUSE galaxies, however, show the coexistence of both positive and negative correlations already at ~ 100 pc scales, and our regulator analysis indicates that the sign is set locally by the *relative amplitude* of variations in $\Sigma_{\rm SFR}$ and the gas-supply-rate surface density (Σ_Φ), rather than by scale alone. If the spaxel-to-spaxel changes in $\Sigma_{\rm SFR}$ are interpreted as being driven primarily by fluctuations in the SFE of individual GMCs, while Σ_Φ is taken as the local analogue of an inflow rate, then Equation 31 is fully consistent with the WL21 framework: positive (negative) correlations emerge where SFE- or SFR-driven (GSR-driven) variability dominates, and both regimes can naturally coexist within the same galaxy at fixed spatial resolution.

On ~kpc scales, Sánchez-Menguiano et al. (2019) used MaNGA spaxels and the O3N2-M13 strong-line calibration to show that the sign and strength of the local $\Delta Z_{\rm gas}$–$\Delta\Sigma_{\rm SFR}$ relation are primarily set by a galaxy's average metallicity: metal-poor systems exhibit the steepest negative slopes, whereas metal-rich systems show a much weaker (or near-zero) trend. They report an inversion at $12 + \log_{10}({\rm O/H}) \simeq 8.5$, which corresponds to $\log_{10}(\Sigma_*) \simeq 7.5$ using our best-fitting rMZR parameters in Equation 14, and is therefore consistent with the inversion point we identify. While our limited sample size ($N = 14$) is insufficient to show an obvious monotonic dependence of the reversal strength on galaxy-averaged metallicity, we note that NGC 4396, which is our most strongly anti-correlated case in Figure 5, also has the lowest integrated oxygen abundance over the **HII**-selected spaxels (by ~0.15 dex relative to the rest, for O3N2-based abundances). In Sánchez-Menguiano et al. (2019)'s interpretation, this behaviour reflects a shift in the origin of the gas fueling star formation: in metal-poor galaxies, localized star-formation enhancements are associated with external accretion of metal-poor gas (strong dilution, hence negative slopes), whereas in metal-rich galaxies the fueling is increasingly dominated by recycled/internal gas (reduced dilution, allowing enrichment to dominate and the relation to flatten or become positive). In the language of our gas-regulator model, these two regimes map naturally onto which term controls the short-timescale departures in $Z_{\rm gas}$. If fluctuations in the gas-supply term dominate, the response is dilution-driven and $\Delta Z_{\rm gas}$ anti-correlates with $\Delta\Sigma_{\rm SFR}$; if instead variability tied to the star-formation/feedback cycle (i.e. $\tau_{\rm dep}$), enrichment can dominate and the relation becomes weak or positive.

Because our galaxies reside in the Virgo Cluster, environmental processes that perturb gas accretion, stripping, and recycling may modulate the local coupling between $Z_{\rm gas}$ and $\Sigma_{\rm SFR}$, plausibly enhancing the role of gas-supply regulation relative to enrichment (Cortese et al. 2021). However, the mass-dependent inversion we observe is not simply a radial-selection effect, because similar behaviour is seen when comparing regions at comparable galactocentric distances across different galaxies. Therefore, we argue that the anti-correlation in low-Σ_* regime is not purely environmentally driven, but instead we treat environment as a plausible contributing factor rather than the primary explanation.

Finally, we note that converting dust-corrected Hα to $\Sigma_{\rm SFR}$ assumes that star formation has been approximately constant over the Hα tracer timescale (a few–$\sim$10 Myr); at $\sim$100 pc this assumption can break down, introducing stochastic (burstiness) fluctuations in the inferred $\Sigma_{\rm SFR}$. In practice, this effect enters our framework as an additional contribution to $\Delta\log_{10}(\Sigma_{\rm SFR})$ in Equation 31, i.e. it is absorbed into the "SFR-driven variability" term ($\delta\log_{10}(\Sigma_{\rm SFR}(t))$). If this extra variance is largely uncorrelated with $Z_{\rm gas}$, it will primarily inflate the scatter and dilute the observed $\Delta Z_{\rm gas}$–$\Delta\Sigma_{\rm SFR}$ trend in regimes where the intrinsic relation is positive; conversely, in regimes where the observed relation is dilution-driven and negative, the same additional excursions in $\Delta\log_{10}(\Sigma_{\rm SFR})$ can make the anti-correlation appear more pronounced. This raises the possibility that part of the inferred inversion could be influenced by Hα-based $\Sigma_{\rm SFR}$ stochasticity; explicitly accounting for such fluctuations would primarily reduce the apparent significance/contrast of the sign-change pattern in Figure 5. However, this cannot be tested with the current data.

4.4 From Spatially-resolved to Integrated Relations

Although this theoretical framework presented so far is written entirely in terms of surface densities, the underlying regulator equations have exactly the same functional form if one replaces $(\Sigma_{\rm g}, \Sigma_*, \Sigma_{\rm SFR}, \Sigma_\Phi)$ by their galaxy-integrated counterparts $(M_{\rm g}, M_*, {\rm SFR}, \Phi)$ and the associated timescales by effective global depletion and inflow timescales. In this sense, Equation 31 can be read as a scale-free statement: at fixed M_*, the sign of the $Z_{\rm gas}$–SFR (anti-)correlation is determined by the comparison of the relative fluctuation amplitudes in SFR and in the inflow rate, or equivalently by whether variations in depletion or inflow timescale dominate. If global fluctuations in SFR (or SFE) are larger than those in inflow, one expects a positive $Z_{\rm gas}$–SFR correlation at fixed M_*; if stochasticity in inflow dominates, dilution produces an anti-correlation. This picture provides a natural physical interpretation for the integrated fundamental metallicity plane/surface found in large samples, where the SFR dependence of metallicity is negative at low stellar mass and commonly observed but weakens and may even change sign at the high-mass end (e.g. Mannucci et al. 2010; Yates et al. 2012; Zahid et al. 2013; Curti et al. 2020). In our view, these global trends reflect the same basic competition between star formation and gas inflow encapsulated in Equation 31, with low-mass galaxies residing in a regime where inflow-driven variability is more significant, and massive systems shifting toward a regime where SFR- or SFE-driven variability becomes comparatively stronger.

Within this picture, the apparent prevalence of anti-correlations in both resolved and integrated studies can be understood, at least in part, as a consequence of how different surveys sample the M_* and/or Σ_* parameter space. Global FMR analyses are dominated by galaxies with $M_* \lesssim 10^{10}, M_\odot$, while more massive systems with $M_* \gtrsim 10^{10.5}, M_\odot$ are rarer and span a narrower range of specific SFR, so the low-mass, inflow-dominated regime contributes a larger fraction of the statistical weight. An analogous effect arises in spatially-resolved work: at kpc-scale resolution, some surveys (e.g. MaNGA and MAGPI) preferentially sample spaxels with $\log_{10}(\Sigma_*/M_\odot, \mathrm{kpc}^{-2}) \lesssim 7.8$ and therefore see predominantly anti-correlated or flat $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ behaviour (Baker et al. 2023; Koller et al. 2024), whereas at $\sim$ 100 pc resolution our MAUVE–MUSE, and possibly MAD data in WL21, ~~maps~~ contain a much larger fraction of spaxels with $\log_{10}(\Sigma_*/M_\odot, \mathrm{kpc}^{-2}) \gtrsim 8$, so the positive branch dominates the ensemble trends. In both the global and resolved cases, the sign flip emerges only when one samples across the transition region (around $\log_{10}\Sigma_* \sim 7.5$–$8.0$ locally and $\log_{10} M_* \sim 10.0$–10.5 globally), and the balance between correlated and anti-correlated regimes is then controlled by the relative number of galaxies/spaxels on either side of this inversion. Furthermore, the more frequent detection of anti-correlation, as well as the weaker and less clear (anti-)correlation signal on global scales than on local scales, may simply raise from an averaging effect, whereby integrating over many spaxels blends regions together with opposite local behaviours. However, (i) what sets the precise location of these transition scales, (ii) whether the local (Σ_*) and global (M_*) inversions are causally linked or merely coincident, and (iii) which physical conditions change across them (e.g. gas fraction, disc stability, outflow efficiency, or environmental processes), remain open questions that we defer to future work.

5 Robustness of Our Findings

In this section, we assess how robust our observed $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ (anti-) correlation is to three key sources of systematics with Figure 6 as a compact visual guide. We first examine the effect of adopting different strong-line abundance prescriptions in Section 5.1, by comparing the four rows of Columns 1 and 3 of Figure 6 (D16, Scal-PG16, O3N2-M13, and O3N2-C20). We then test the sensitivity to star-forming spaxel selection in Section 5.2, by contrasting the conservative **HII** results (Columns 1 and 3) with the more inclusive **HII+Comp** (the composite region in the N II BPT diagram is also considered as "star-forming") results (Columns 2 and 4; the fraction of spaxels used in the map become 45.89%). Finally, we investigate the special case of the metallicity outlier NGC 4383 in Section 5.3, highlighted by dotted contours or curves in each panel of Figure 6 relative to the solid loci of the remaining galaxies. The broader purpose of these checks is to emphasise that, although calibration choice, excitation mixing, environment, and mass-dependent sampling can modulate the apparent scatter and the detailed slope/normalization of rMZR in individual regimes, they do not alter the underlying physical interpretation implied by Equation 31, namely that the sign of the local residual correlation is set by the competition between the relative fluctuations of star formation and gas supply, $\delta\log_{10}\Sigma_{\rm SFR}$ versus $\delta\log_{10}\Sigma_\Phi$, or equivalently by the balance between gas supply and gas depletion timescales,$\delta\log_{10}\tau_\Phi$ and $\delta\log_{10}\tau_{\rm dep}$.

5.1 Sensitivity to Metallicity Calibration

A well-known limitation of any resolved metallicity analysis is that strong-line prescriptions are not anchored to a unique absolute scale. Different calibrations rely on different reference samples and methodologies—direct electron-temperature (T_e) measurements, photoionization model grids, or hybrid/stacked approaches—and they respond differently to variations in excitation, ionization parameter, DIG, and N/O abundance. Consequently, systematic offsets of order $\sim$ 0.1–0.3 dex (and in extreme cases ~~larger~~ up to $\sim$ 0.5 dex) between calibrations applied to the same spectra are well documented, and can alter both the normalization and the slopes of resolved relations (e.g. Kewley & Ellison 2008; Kewley et al. 2019; Maiolino & Mannucci 2019). These long-standing uncertainties motivate our use of four widely adopted diagnostics spanning empirical T_e-calibrated and model-based families: D16, Scal-PG16, O3N2-M13, and O3N2-C20.

The first column of Figure 6 shows that, despite calibration-dependent zero-points, the overall rMZR morphology is broadly stable across all prescriptions: $Z_{\rm gas}$ increases steeply with Σ_* at low surface densities and flattens toward a saturation-like regime at high Σ_*, in line with previous resolved surveys. However, the secondary dependency on $\Sigma_{\rm SFR}$ is not equally coherent for every diagnostic. For both O3N2-based calibrations, the $\Sigma_{\rm SFR}$-stratified contours display a clear and systematic reversal with Σ_*: at low Σ_*, higher $\Sigma_{\rm SFR}$ preferentially occupies lower $Z_{\rm gas}$, while at high Σ_* the ordering flips such that higher $\Sigma_{\rm SFR}$ corresponds to higher $Z_{\rm gas}$. This behaviour is reflected in the ensemble Spearman trends (third column), which show a coherent transition from $\rho < 0$ at low Σ_* to $\rho > 0$ at intermediate/high Σ_* for both O3N2 prescriptions (with galaxy-to-galaxy scatter but a consistent global trend). By contrast, the D16 and Scal-PG16 prescriptions exhibit substantially weaker and more heterogeneous $\Sigma_{\rm SFR}$ stratification in the H II-only sample. The D16 ensemble ρ trend remains mildly positive over most of the Σ_* range; while Scal-PG16 shows a stronger dependence on Σ_* in the ensemble statistic, but with opposite-sign behaviour compared to O3N2-M13 and O3N2-C20. Therefore, while the rMZR curvature itself is stable, the inferred $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ coupling (and any apparent sign change) is diagnostic-dependent.

The comparison also reveals systematic differences between calibration families. In our sample, indicators tied more closely to T_e-based anchors (e.g. Scal-PG16 and both O3N2 calibrations) show tighter rMZR trends with metallicity dispersions of $\sim 0.1-0.15$ dex at fixed Σ_* and $\Sigma_{\rm SFR}$, whereas the model-based D16 prescription produces visibly larger scatter ($\sim$ 0.2 – 0.25 dex) and spans a broader metallicity range ($\sim$ 0.4 – 0.5 dex wider) across the same parameter space. This behaviour is consistent with PHANGS–MUSE (Kreckel et al. 2019), which shows that calibrations tied to photoionization models or to a small number of line ratios are more susceptible to local variations in ionization parameter and abundance ratios, leading to increased spaxel-to-spaxel dispersion and systematic offsets relative to T_e-anchored scales. Within the O3N2 family itself, the two prescriptions are highly consistent in their trends; the main difference is a nearly uniform offset, with O3N2-C20 yielding metallicities systematically higher than O3N2-M13 by $\sim$ 0.1–0.2 dex.

While both O3N2-based calibrations recover a similar inversion, the presence and coherence of the sign change are not universal across all prescriptions. The D16 Spearman trends are dominated positive ρ over most of the Σ_* range, mirroring the predominantly positive resolved correlation reported by WL21 on $\sim$ 100 pc scales. This suggests that the *choice of calibration* can influence whether a given data set appears dominated by correlated or anti-correlated

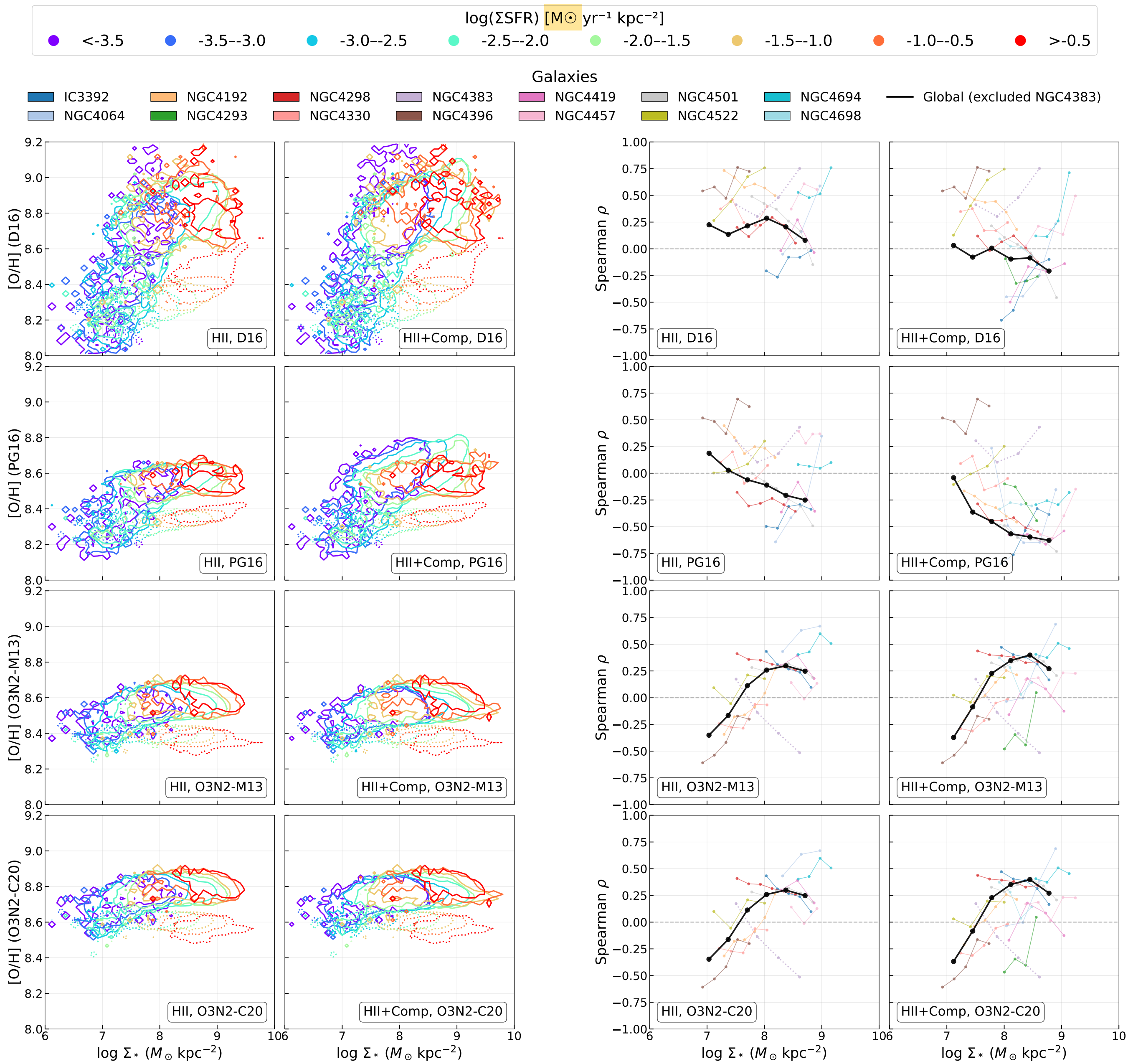

Figure 6. Comparison of the rMZR (left two columns) and the mass-dependent Spearman rank correlation coefficient ρ between Δ[O/H] and $\Delta \log_{10} \Sigma_{\rm SFR}$ (right two columns) for four gas-phase metallicity calibrations: D16 (top row), Scal-PG16 (second row), O3N2-M13 (third row), and O3N2-C20 (bottom row). Columns 1 and 3 use only **HII**-classified star-forming spaxels, whereas Columns 2 and 4 use the more inclusive **HII+Comp** selection (see the BPT demarcation in Figure 1). The rMZR panels are colour-coded by $\log_{10} \Sigma_{\rm SFR}$ as in Figure 3. The Spearman panels show $\rho(\Sigma_*)$ measured in discrete Σ_* bins for individual galaxies (coloured curves) and for the ensemble sample (black curve; excluding NGC 4383), following Figure 5. The parameter spaces occupied by NGC 4383 are indicated by dotted contours/curves in each corresponding panel.

regimes, even when the underlying spaxel population is the same. Moreover, at the high-Σ_* end, the O3N2 prescriptions favour a positive residual coupling, whereas D16 and Scal-PG16 lean toward weaker or even negative behaviour. Similar calibration-dependent qualitative differences have been noted in other MUSE-based surveys, where O3N2- and N2-type indicators can yield different resolved slopes from S- or N2S2-based prescriptions (e.g. Kreckel et al. 2019). We therefore caution that the detailed mass dependence of the $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ relation can diminish or even reverse under different metallicity calibrations.

In general, both O3N2 prescriptions are also the ones that most cleanly delineate the specific behaviour of interest here: they show a coherent transition from an anti-correlation between $Z_{\rm gas}$ and $\Sigma_{\rm SFR}$ at low Σ_* to a positive coupling at higher Σ_*. In our resolved sample, this inversion occurs at $\log_{10}(\Sigma_*/M_\odot\,{\rm kpc}^{-2}) \approx 7.5$–$8.0$. From a physical point of view, such a transition is qualitatively expected in gas-regulator frameworks: in low-Σ_* (typically lower-metallicity) regimes, fluctuations in gas supply can dominate the short-timescale response and produce dilution-driven anti-correlations, whereas in higher-Σ_* regimes the response can be increasingly enrichment-

/consumption-dominated, yielding a positive residual coupling. This qualitative pattern is consistent with the global, integrated picture established using SDSS data, whose oxygen abundances derived by both strong line diagnostics and grids of photoionization models reveal a similar reversal, negative at low masses and positive at high masses, with a transition point around a characteristic mass of $\log_{10}(M_*/M_\odot) \approx 10.0 - 10.5$ (see Figure 1 of Yates et al. 2012 and Figure 4 of Zahid et al. 2013). Similarly, although the signal is not strong, Curti et al. (2020) also present such a reversal of $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ correlation at the high-mass end in their Figure 6 and 7.

In addition, we emphasize that the inferred sign structure is not calibration-independent. For example, D16 implies that the residual coupling is predominantly positive across most of the Σ_* range, which would suggest enrichment-dominated behaviour even in the lowest-Σ_* regime probed here. Given that our galaxy sample spans a limited range in global stellar mass, we cannot rule out that more low-mass systems would strengthen the dilution-dominated branch; moreover, D16 relies on ratios involving [N II] and [S II], which can be particularly sensitive to variations in N/O and to DIG/shock contributions at ∼100 pc resolution, potentially biasing $\rho(\Sigma_*)$ toward positive values and/or increasing the scatter. In contrast, Scal-PG16 yields a less coherent $\rho(\Sigma_*)$ behaviour and can imply comparatively stronger anti-correlation at high Σ_*, which is difficult to reconcile with the integrated reversal and basic regulator expectations, suggesting that systematic sensitivity to local excitation/line-ratio mixing may dominate in this calibration. We therefore adopt the O3N2 prescriptions as our fiducial description of the inversion while treating D16 and Scal-PG16 primarily as systematic cross-checks on calibration-dependent behaviour.

5.2 Sensitivity to the Star-forming Spaxel Definition

Our baseline analysis relies on a conservative **HII**-only mask, in which spaxels are required to fall within the **HII** locus of both the [N II] and [S II] BPT diagrams with robust line detections and uncertainties that remain fully inside the same excitation class. This stringent selection is designed to minimise contamination from shocks, AGN-like excitation, and DIG, all of which can bias strong-line metallicity indicators and Hα-based $\Sigma_{\rm SFR}$ estimates. However, MAUVE–MUSE also provides a more inclusive "star-forming" mask that merges **HII** and composite regions (**HII+Comp**), motivated by the expectation that composite spaxels are still largely powered by O/B stars' photonization but may include a non-negligible contribution from DIG or harder radiation fields. We therefore repeat the full rMZR and offset-correlation analysis using **HII+Comp** spaxels to assess the sensitivity of our conclusions to excitation masking.

The comparison is summarised in Figure 6 as well, where Columns 1 and 3 use the conservative **HII** selection, and Columns 2 and 4 adopt **HII+Comp**. Including composite spaxels expands the sampled parameter space, especially toward high Σ_* where inner discs and circumnuclear regions are more likely to host mixed excitation and more complex phyiscs. For the two O3N2-based prescriptions (O3N2-M13 and O3N2-C20), both the rMZR colour ordering and the mass-dependent Spearman trends are essentially unchanged when composites are added. The ensemble $\rho(\Sigma_*)$ curves retain the same negative-to-positive transition at $\log_{10}(\Sigma_*/M_\odot\,{\rm kpc}^{-2}) \sim 7.5$–$8.0$, and the scatter around the media rMZR locus increases only marginally. This stability indicates that our main result is not driven by a narrow excitation-class definition, and that O3N2 indicators are comparatively resilient to moderate DIG/composite admixture in the MAUVE–MUSE data.

By contrast, the N2S2- and S-calibrations show a stronger response to including composites. In particular, the Scal-PG16 rMZR exhibits a noticeable increase in scatter at the high-Σ_* end under the **HII+Comp** mask, consistent with the known sensitivity of S2- and N2-based ratios to harder ionizing spectra and DIG-enhanced low-ionization lines. The corresponding $\rho(\Sigma_*)$ trends also shift: for D16 the ensemble Spearman curve becomes dominated by weak or negative correlations over a broader Σ_* range, and some individual galaxies show reduced significance or a delayed zero-crossing; while for Scal-PG16 the ensemble Spearman trend become completely negative without no inversion point anymore. These changes underline that the N2S2- and S-calibrations (i.e., D16 and Scal-PG16 here) leveraging [N II] and [S II] can be more susceptible to excitation mixing, and they reinforce the need to treat excitation masking and calibration choice as coupled systematics when interpreting resolved $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ relations.

Crucially, despite these quantitative shifts, the qualitative inversion of the residual $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ coupling persists in the ensemble sample for both O3N2-M13 and O3N2-C20. The low-Σ_* anti-correlation branch and the high-Σ_* positive branch remain present, demonstrating that the sign flip is not an artefact of excluding composite spaxels when using O3N2 prescriptions.

5.3 NGC4383 – the Metallicity Outlier

NGC 4383 is a clear metallicity outlier within the MAUVE–MUSE sample. To illustrate its behaviour, we reintroduce its spaxels in Figure 6 as dotted contours (rMZR panels) and dotted light-mauve curves (Spearman panels), while keeping the ensemble Spearman trend (black curve) computed from the other 13 galaxies only. From the rMZR comparison (first two columns), NGC 4383 remains systematically offset to lower $Z_{\rm gas}$ at fixed Σ_* by at least 0.2 dex under all four metallicity calibrations. The persistence of this depression across prescriptions and excitation selections indicates that it is not produced by calibration-specific systematics or by the choice of star-forming mask, motivating our designation of NGC 4383 as a metallicity outlier.

The Spearman trend of NGC 4383 adds a second, more striking peculiarity. Unlike the other galaxies, NGC 4383 shows a mass-dependent correlation pattern that is nearly the mirror image of the ensemble relation: wherever the combined MAUVE–MUSE behaviour is positive (negative), the NGC 4383 curve tends to be negative (positive), especially in O3N2 indicators. In other words, the sign of the residual $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ coupling in NGC 4383 appears to reverse relative to the dominant local regime traced by the rest of the sample. If included in the stack, this anti-aligned behaviour would disproportionately populate the low-$Z_{\rm gas}$ tail and partially cancel the mass-ordered trend, artificially steepening the low-Σ_* rMZR branch and shifting $\rho(\Sigma_*)$ toward the non-correlation case. We therefore exclude NGC 4383 from all ensemble rMZR and correlation measurements in the main analysis part of the paper, and investigate it separately here. Such abnormal behaviour of NGC 4383 in Figure 6 under the O3N2 indicators implies that, in contrast to the other 13 MAUVE–MUSE galaxies, NGC 4383 is dominated by GSR-driven variability in the $\log_{10}(\Sigma_*/(M_\odot\,{\rm kpc}^{-2})) \gtrsim 7.8$ regime.

A plausible physical explanation for both its depressed metallicity scale and its opposite-sign residual coupling is a substantial reservoir of (re-)accreted, relatively metal-poor gas that enhances gas-supply variability and dilutes $Z_{\rm gas}$ at fixed Σ_*. This interpretation is consistent with the kinematic anomalies reported by Chung et al. (2009) and with the view that NGC 4383 has recently acquired a large gas reservoir (Cortese et al. 2026). NGC 4383 has been identified as a system with strong feedback activity in the MAUVE–MUSE context

(Watts et al. 2024), and such activity may further perturb the local balance between enrichment and gas supply that sets the sign of the $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ relation. However, especially at the low Σ_* regime, part of the ionized emission may be dominated by extraplanar/outflowing gas rather than classical disc H II regions, in which case strong-line calibrations and Hα-based $\Sigma_{\rm SFR}$ become less reliable tracers of in-situ star formation. NGC 4383 therefore provides an existence proof that the commonly observed rMZR and the local $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ coupling can appear to break down when the ionized emission is strongly affected by non-disc gas (e.g. outflows/extraplanar components) and/or when the system is subject to strong, recent gas-supply perturbations.

6 Conclusions

In this paper, we have analysed 14 Virgo spiral galaxies from the MAUVE–MUSE second internal release to revisit the resolved mass–metallicity relation and its secondary dependence on star formation rate surface density on ~ 100 pc scales. By combining high-quality IFS data with a spatially-resolved regulator framework, we have characterised where and why the local gas-phase metallicity to star formation rate (SFR) surface density ($Z_{\rm gas}$–$\Sigma_{\rm SFR}$) relation (and also its residual coupling) changes sign. Our main conclusions are as follows.

(i) Our MAUVE–MUSE galaxies follow a well-defined resolved mass–metallicity relation that is consistent with previous IFS surveys. Using the O3N2-M13 calibration as our fiducial metallicity scale, the resolved mass metallicity relation (rMZR) rises steeply at low stellar mass surface density (Σ_*) and flattens toward high Σ_*, in line with earlier work on kpc-scale data. At fixed Σ_*, low-Σ_* regions tend to show lower $Z_{\rm gas}$ at higher $\Sigma_{\rm SFR}$, whereas high-Σ_* regions show the opposite ordering (see Figure 3).

(ii) After removing the primary rMZR and resolved star formation main sequence (rSFMS) trends, the residual coupling between metallicity and SFR surface density exhibits a robust mass-dependent inversion. The Spearman rank correlation coefficient between Δ[O/H] and $\Delta\log_{10}\Sigma_{\rm SFR}$ transitions smoothly from $\rho < 0$ at $\log_{10}(\Sigma_*/M_\odot\,{\rm kpc}^{-2}) \lesssim 7.5$ to $\rho > 0$ at $\log_{10}(\Sigma_*/M_\odot\,{\rm kpc}^{-2}) \gtrsim 8.0$, with a zero-crossing in between (see Figure 5). Notably, NGC 4192 spans a broad Σ_* range that straddles the inversion point and exhibits a clear internal sign reversal, rather than simply populating one branch or the other. This suggests that the transition is driven by an intrinsic physical mechanism operating within individual galaxies, rather than arising as an artifact of stacking different galaxy populations. Local Moran-like correlation maps further show that, once Σ_* trends are removed, correlated and anti-correlated H II regions can coexist within the same galaxy, and the familiar mass-dependent behaviour emerges only as an ensemble property.

(iii) To interpret these patterns, we formulated a spatially-resolved gas–regulator model in surface densities and identified an "instantaneous target metallicity" ($Z^\dagger$) that drives the evolution of the observed real-time metallicity (Z). Analysis around metallicity extrema and toward/away from them leads to a simple condition (see Equation 31): at fixed Σ_*, the sign of the local $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ (anti-)correlation is controlled by the relative fluctuation amplitudes of star formation and gas supply. Positive correlations arise where variability of SFR or star formation efficiency (SFE) dominates (short, strongly varying depletion times at nearly steady gas supply), while negative correlations arise where variability of gas supply rate (GSR) dominates and dilution drives $Z_{\rm gas}$ opposite to $\Sigma_{\rm SFR}$. Our MAUVE–MUSE trends are therefore naturally understood as the outcome of a competition between SFR-driven and GSR-driven fluctuations that vary systematically with Σ_*. The physical drivers of these variabilities will be explored in the future work.

(iv) We argued that the same gas-regulator physics can be extrapolated from spatially-resolved to integrated galaxy-wide relations (Section 4.4). Replacing surface densities by their global analogues, Equation 31 implies that the sign of the integrated $Z_{\rm gas}$–SFR dependence at fixed M_* is likewise set by whether fluctuations in SFR (or SFE) outweigh those in the inflow rate. In this view, the negative SFR dependence of metallicity commonly seen at low stellar masses and the weaker or reversed dependence at the high-mass end in global FMR analyses are two manifestations of the same underlying competition between depletion and inflow timescales. The apparent prevalence of anti-correlation, and the weaker and less clear direct correlation signal on global scales than on local scales, can be understood as a consequence of parameter-space sampling and spatial averaging: many surveys predominantly probe galaxies or spaxels on the low-mass, inflow-dominated side of the inversion and then integrate over regions with mixed local behaviour.

(v) Finally, we tested the robustness of these conclusions against three key systematics (see Figure 6). First, adopting four widely used metallicity diagnostics (D16, Scal-PG16, O3N2-M13, and O3N2-C20), we find that the overall rMZR curvature is stable across all calibrations. However, the mass-dependent sign flip in the residual $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ coupling can change according to the choice of diagnostic. The O3N2-based indicators (O3N2-M13 and O3N2-C20) show the clearest and most consistent transition from anti-correlation at low Σ_* to correlation at high Σ_*. By contrast, the Scal-PG16 prescription exhibits a completely opposite trend, while the D16 calibration remains predominantly positively correlated over the full Σ_* range. We therefore conclude that, although the resolved metallicity scaling relations are qualitatively similar across diagnostics, the presence and sign of the inferred $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ inversion are calibration-dependent. Second, relaxing the star-forming mask from a conservative **HII**-only selection to an inclusive **HII+Comp** mask increases the scatter and modulates the detailed $\rho(\Sigma_*)$ trends, with the strongest changes occurring for the N2S2- and S-based diagnostics (D16 and Scal-PG16). For the O3N2-based calibrations (O3N2-M13 and O3N2-C20), the mass-dependent inversion remains identifiable. In contrast, under the Scal-PG16 prescription the coupling becomes predominantly negative across the full Σ_* range, while D16 shows no clear monotonic correlation, with the stacking Spearman trend fluctuating around zero. Third, we showed that NGC 4383 is a genuine metallicity outlier: it lies ~ 0.1–0.2 dex below the rMZR locus and exhibits an almost mirror-image residual correlation pattern relative to the stack (Section 5.3).

In summary, these results suggest that the spatially resolved $Z_{\rm gas}$–$\Sigma_{\rm SFR}$ relation is not governed by a single universal trend, but by a balance between star-formation-driven and gas-supply-driven variability that depends on local mass surface density, environment, and feedback. Our MAUVE–MUSE data show that both correlated and anti-correlated regimes naturally coexist within the same discs at ~ 100 pc resolution, and that the familiar mass-dependent inversion emerges as a population-level imprint of this underlying competition. Future work combining MAUVE–MUSE data with resolved H I and CO observations, as well as higher-redshift IFS surveys will be crucial for constraining gas supply rate surface density (Σ_Φ) and its timescale more directly, and for testing whether the inversion scales identified here are universal or themselves evolve with redshift, environment, and galaxy assembly history.

Software

This research made use of astropy (Astropy Collaboration et al. 2022, https://www.astropy.org/), numpy (Harris et al. 2020, https://numpy.org), matplotlib (Hunter 2007, https://matplotlib.org/), scipy (Virtanen et al. 2020, https://scipy.org/), speclite (Kirkby et al. 2023, https://speclite.readthedocs.io/en/latest/) and nGIST (Fraser-McKelvie et al. 2025, https://geckos-survey.github.io/gist-documentation/).

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Data Availability

The full set of MAUVE–MUSE datacubes and associated nGIST value-added products will be released as part of the main public data release once the survey is completed (anticipated in 2027). Requests for access prior to the public release should be directed to the corresponding author. Readers interested in joining the MAUVE collaboration and accessing the full dataset are encouraged to consult the team page on the MAUVE survey website. The analysis pipeline used in this work is publicly available at https://github.com/Rongjun-ANU/MAUVE_MUSE_Zgas_SigmaSFR.

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